An optimisation approach for supply chain procurement transport operational planning with 3D full-truckload constraints and returnable transport items

In procurement transport planning with returnable transport items (RTIs) — foldable containers that must be returned empty — the delivery and the return flow are each other's supply. We formulate 3D-SCPTOP, a mixed integer programme joining the transport and inventory decisions to a full-truckload condition stated as a 3D loading problem: RTIs form vertical stacks under a height budget and a per-stack crush limit, whose footprints must be placed on the trailer floor without overlap. The formulation is written over trucks, so it is block-angular over (direction, truck, period); deleting that quantifier yields the pricing problem of a Dantzig–Wolfe reformulation whose master carries *no* loading configuration. We solve it with BP-3D-SCPTOP: branch and price over the fleet vector, with a flow-first relax-and-fix construction as primal heuristic and a lower bound certified on the volumetric relaxation, valid for the 3D problem by inclusion. The branching hierarchy is finite and complete, so the procedure is exact given compute; what it reports here is an incumbent verified truck by truck against its own layout, a certified bound, and no proof of optimality. On thirty-six generated instances it returns a verified plan on thirty-one, at a mean certified gap of 8.8% over thirty of them; on four size-ladder instances a compact 3D formulation returns a plan only on the smallest, of five periods. Of the 724 truck loads a volumetric capacity model dispatches on those instances, 501 (69.2%) are *proved* impossible to place in a trailer.

Keywords— Procurement transport planning, inventory, returnable transport items, full-truckload, 3D container loading, branch and price

1 Introduction

Taking transport, inventory and production decisions jointly improves supply chain performance (Stadtler, Kilger, and Meyr 2015; Díaz-Madroñero, Mula, and Peidro 2017). Responsibility for transport planning usually lies with the supplier (Fleischmann, Kopfer, and Sürie 2015), but in the automotive industry the manufacturer frequently plans its own inbound transport, and it does so around an asset material requirements planning does not represent: the returnable transport item (RTI), the tertiary packaging in which parts move. The plant returns folded empty RTIs and the supplier sends them back erected and full, so each flow is the other's supply and neither can be planned alone. Algorithm-based planning outperforms a planner's intuition here (Díaz-Madroñero, Peidro, and Mula 2014), most where intuition is weakest: deciding how many trucks to dispatch when whether a given set of containers fits inside one is itself a combinatorial question.

That last question is the one the literature does not ask. Transport capacity in procurement transport planning is a scalar — freight volume, or a count of loaded units (Hashemi-Amiri, Ghorbani, and Ji 2023; Díaz-Madroñero, Peidro, and Mula 2014) — but containers are not fluid: they stack in whole layers under a height budget and a crush limit, and their footprints tile a floor in whole rectangles. A plan drawn against a volumetric capacity can therefore prescribe loads that no arrangement of the trailer admits, and the shortfall is discovered on the loading dock rather than in the model. Section 9.5 measures how often, by placeability proof rather than by argument.

This paper makes four contributions, listed in the order of the evidence behind them.

1. A mixed integer formulation, 3D-SCPTOP (Section 4), joining procurement transport, RTI closed-loop and inventory decisions to a full-truckload condition stated as a stack-based 3D loading problem, with the per-stack crush cap evaluated on the product mix actually loaded in each slot rather than on a worst case for its container family.

2. A Dantzig–Wolfe reformulation (Dantzig and Wolfe 1960) and a branch-and-price procedure, BP-3D-SCPTOP (Sections 5–8), derived structurally rather than by analogy: deleting the quantifier over (direction, truck, period) from the block rows *is* the pricing problem. The procedure branches over the fleet vector with a flow-first relax-and-fix construction as primal heuristic, and its lower bounds rest on one proposition: a certified bound of the volumetric relaxation is valid for the 3D problem.
3. Accuracy against a reference optimum (Section 9.2): on the volumetric problem, where a compact formulation solves to proved optimality, the procedure's plan and its certified bound bracket that optimum within about a hundredth of a percent.
4. A measurement of what a volumetric transport capacity costs (Section 9.5), with the mechanism behind it and a test on the industrial container catalogue that a planner can run before solving anything.

The branching hierarchy is finite and exhaustive and every prune is a proof, so BP-3D-SCPTOP is an exact procedure run under a budget rather than a heuristic; on the sizes reported here it returns a verified plan and a certified bound, and proves no optimum.

2 Related work

Procurement transport with inventory decisions. (Berman and Wang 2006) and (Sancak and Salman 2011) price full and less-than-truckload service against inventory under a volumetric capacity; (Schöneberg, Koberstein, and Suhl 2010) and (Schöneberg, Koberstein, and Suhl 2013) do so for multi-supplier inbound networks; (Díaz-Madroñero, Mula, and Peidro 2014) formulate the operational single-supplier full-truckload problem that the benchmark of Appendix A extends, with capacity as a maximum container count; (Hein and Almeder 2016), (Díaz-Madroñero, Mula, and Peidro 2017), (Meyer and Amberg 2018), (Chitsaz, Cordeau, and Jans 2019), (Baller et al. 2022) and (Hashemi-Amiri, Ghorbani, and Ji 2023) extend the setting to milk-runs, routing and uncertainty. In every case the trailer is a scalar: a volume, a weight, or a number of units. Table 1 positions these models on the features that matter here, and in no prior row is the capacity a condition on placement.

RTIs. The closed loop itself is modelled at a tactical or network level. (Soysal et al. 2023) plan vendor-managed inventory of agro-food crates with forward and reverse flows; (Guo et al. 2024) optimise a two-tier container management network with return penalties; (Zhang et al. 2024) formulate a perishable inventory-routing problem with several RTI types. These carry the reverse flow but represent the vehicle as a capacity, so a folded container's occupying a different volume from an erected one changes a coefficient and not the structure of the problem.

Container loading. The loading side is surveyed by the typology of (Wäscher, Haußner, and Schumann 2007), the review of practically relevant constraints of (Bortfeldt and Wäscher 2013) and the exact-method survey of (Silva, Toffolo, and Wauters 2019). Our loading requirements — orientation, stacking with load bearing, weight limits — are those studied as side constraints of a single-container problem by (Junqueira, Morabito, and Sato Yamashita 2012) and (Alonso, Alvarez-Valdes, and Parreño 2020). The nearest methodological precedent is (Mahvash, Awasthi, and Chauhan 2018), who solve three-dimensional bin packing by column generation in which a column is a packing configuration of one bin: we take the same unit of decision, but the master it feeds is a multi-period transport and inventory problem rather than a set-covering model, so the columns of a direction are generated once and priced against duals that move with the inventory balance.

3 Problem description

A plant assembles products supplied by a single supplier over a finite horizon. Every product travels in exactly one type of RTI: a rigid container, *erected* while it carries product and *folded* when empty, so one physical container occupies two different volumes depending on which way it is moving. The supplier packs its production into erected RTIs and ships them to the plant; the plant consumes the product, folds the containers and returns them. Both legs are run with identical trailers, every dispatch costs the same whatever it carries, and a trailer may be dispatched only if it is full. A dispatch arrives lt periods after it leaves and is available at neither end in between, and each node must end every period holding enough stock to cover the next periods of its own activity. What has to be decided is, per direction and period, how many trucks

Table 1: Procurement transport models with inventory decisions, on the features that separate them from 3D-SCPTOP. *Config.*: SS single source and single destination, MS multi-supplier, MM multi-echelon. *Level*: O operational, T tactical. *Mode*: FTL full-truckload, LTL less-than-truckload, MR milk-run, CES courier express. *Capacity*: V volumetric, W weight, UH a count of loaded units. *RTIs*: S one type, M several, F foldable, NF not. *Framework*: (F)(MO)(I)LP linear programming variants, CCP chance-constrained. A dash is a feature the model does not carry.

Article	Config.	Level	Mode	Capacity	Tr. cost	Inv. cap.	RTIs	Inv. cost / safety stock	Framework
(Berman and Wang 2006)	MM	T	FTL | LTL	V	X	—	—	X / —	MINLP
(Sancak and Salman 2011)	—	O	LTL | FTL	V	X	—	—	X / X	MILP
(Schöneberg, Koberstein, and Suhl 2010)	MS	T	FTL | LTL	V	X	X	—	X / —	MILP
(Schöneberg, Koberstein, and Suhl 2013)	MS	O	FTL | LTL	V	X	X	—	X / —	MILP
(Díaz-Madroñero, Mula, and Peidro 2014)	SS	O	FTL	UH	X	—	—	X / X	FMOILP
(Hein and Almeder 2016)	MS	O	MR	V	X	—	—	X / —	MILP
(Díaz-Madroñero, Mula, and Peidro 2017)	MS	T	FTL | LTL | MR	V	X	X	—	X / X	MILP
(Meyer and Amberg 2018)	MS	T, O	MR | FTL	V	X	X	—	X / —	MILP
(Chitsaz, Cordeau, and Jans 2019)	MS	O, T	MR	V	X	X	—	X / —	MILP
(Baller et al. 2022)	MS	O	MR | FTL | CES	V | W	X	X	—	— / X	MILP
(Hashemi-Amiri, Ghorbani, and Ji 2023)	MS	T	MR	V	X	—	—	— / —	CCP
(Guo et al. 2024)	MM	T, O	FTL	UH	X	X	S | NF	X / —	MINLP
(Zhang et al. 2024)	MS	T, O	MR	UH	X	X	M | NF	X / —	ILP
3D-SCPTOP	SS	O	FTL	**3D loading**	X	X	M | F	X / X	MILP

go, what each carries and what consequently sits in stock; the cost is the number of dispatches plus the inventory held.

Two features separate this from a transport-and-inventory model. The two flows are coupled through the containers themselves — the plant can return only empties it has emptied and the supplier can fill only containers it has received — so a shortage on the return leg propagates into the delivery leg one lead time later. And the admissibility of a truck load is a three-dimensional packing condition rather than a capacity inequality: a load whose volume and weight both fit may still have no arrangement inside the trailer. Loading is represented through *stacks* of whole homogeneous layers rather than through the placement of individual RTIs, which makes the stacks self-supporting, so no vertical-stability constraint is needed.

4 The 3D-SCPTOP formulation

The model is written over *trucks*. A dispatched truck designs its own layout — the stack types, orientations, positions and layer compositions of its slots are variables of that truck and that period, and its load is assigned to those slots in the same period — so there is no catalogue of loading configurations. That is the decision Section 5 rests on: every row below carries either the triple (g, k, t) or none of it, so the model is block-angular over $G \times K \times T$ and the pricing problem of the Dantzig–Wolfe reformulation is obtained by deleting that quantifier and nothing else.

Throughout, $g \in G = \{s, p\}$ indexes nodes and directions (named by destination), $n \in N$ products, $r \in R$ RTI types, $h \in H$ stack types, $i, j \in I$ slot positions, $k \in K$ trucks and $t \in T$ periods, with $\rho(n)$ the RTI type product n travels in and $M(r) = \{n : \rho(n) = r\}$; variables whose direction is fixed by definition omit g. Table 2 is the nomenclature.

Table 2: Nomenclature. The pricing problem of Section 7 is one block of this model with the index (g, k, t) deleted, so it reuses these symbols with that index dropped; the three variables it adds are listed last.

Indices and sets	
G	Supplier and plant nodes, $G = \{s, p\}$; $g = s$ is the folded direction (plant to supplier), $g = p$ the full one.
N, R, H	Products, RTI types, stack types.
I	Stack-slot indices on a truck floor, bounded by (46).
K, T	Trucks available per direction and period; planning periods.
$\rho(n)$, $M(r)$	The RTI type product n travels in; the products type r carries, $M(r) = \{n : \rho(n) = r\}$.
$R(h)$	RTI types compatible with stack type h.
Parameters	
cK	Cost of dispatching one truck.
lt, SC	Lead time, $lt \geq 1$; coverage window, in periods.

i_{gn}, ie_{gr}	Initial inventory at g of full RTIs of n, of folded RTIs of r.
ic_{gn}	Inventory holding cost of n at g.
d_{gnt}	Production plan for n at g in t: filled at s, consumed at p.
a_{nt}, ae_{rt}	Pipeline arrivals at p, respectively s, in $t \leq lt$.
ss_{gnt}, sse_{gtr}	Coverage stock of full, respectively folded, RTIs (20).
w_n, w^{f}_r	Weight of a full RTI carrying n; of a folded RTI of type r. Both > 0.
pK^w, $W^{\max}$	Truck payload; per-stack crush-limit cap on the *total* weight of one stack, over all its layers.
pK, qK, rK	Truck floor length, floor width, inner height; $\Gamma_K = pK \cdot qK \cdot rK$.
α	Minimum truck utilisation, on whichever of volume or payload binds, $0 < \alpha \leq 1$.
p_h, q_h	Footprint length and width of stack type h.
r_{gr}, a_r, γ_{gr}	Height of one RTI of type r at node g (full at p, folded at s); its footprint area; its volume $\gamma_{gr} = a_r r_{gr}$.
n_{hr}	RTIs of type $r \in R(h)$ per layer of stack type h; one layer tiles the footprint, $n_{hr} a_r = p_h q_h$.
$\hat{w}_{gr}$	Weight per RTI in the layer-capacity form of the utilisation floor: $\hat{w}_{sr} = w^{\mathrm{f}}_r$, $\hat{w}_{pr} = \max_{n \in M(r)} w_n$.
η^K, η^I	Weights of the transport and inventory objectives; both 1 throughout.
M^x, M^y	Big-M constants of the non-overlap rows, $M^x = pK$, $M^y = qK$.
$\bar{u}^V_n$, $\bar{u}^{\mathrm{e}}_r$	$\lfloor W^{\max}/w_n \rfloor$ full RTIs of n per slot; $\lfloor pK^w/w^{\mathrm{f}}_r \rfloor$ folded RTIs of r per truck.
Decision variables — the truck and its layout	
Z_{gkt}	$\in \{0,1\}$. Truck k of direction g in t is configured, i.e. carries a layout.
Y_{gkt}	$\in \{0,1\}$. That truck is dispatched.
θ_{gkt}	$\in \{0,1\}$. Selects the resource on which its utilisation floor is required: volume if 1, payload if 0.
X_{gihkt}	$\in \{0,1\}$. Slot i holds a stack of type h.
L_{gihktr}	$\in \mathbb{Z}_{\geq 0}$. Layers of RTI type $r \in R(h)$ in slot i.
LX_{gihkt}, LY_{gihkt}	$\in \{0,1\}$. Stack h's length lies along the x-, respectively y-axis in slot i.
PX_{gikt}, PY_{gikt}	≥ 0. Bottom-left coordinates of slot i on the floor.
A, B, C, D (on $gijkt$)	$\in \{0,1\}$. Which of the four separating half-planes resolves the overlap of slots i and j.
Decision variables — load and inventory	
V_{kint}	$\in \mathbb{Z}_{\geq 0}$. Full RTIs of n in slot i of plant-bound truck k in t.
V^{e}_{krt}	$\in \mathbb{Z}_{\geq 0}$. Folded RTIs of type r on supplier-bound truck k in t.
I_{gnt}, Ie_{gtr}	$\in \mathbb{Z}_{\geq 0}$. End-of-period inventory of full RTIs of n, of folded RTIs of r, at g.
Derived quantities (abbreviations, (1)–(5))	
$\overline{V}_{nt}$, $\overline{V^{\mathrm{e}}}_{rt}$	Full RTIs of n delivered to the plant in t; folded RTIs of r returned to the supplier.
$\overline{V}_{knt}$	Full RTIs of n on plant-bound truck k in t.
$\widehat{X}_{gikt}$, $\overline{L}_{ghktr}$	Slot i is used; layers of type r under stack type h on that truck, over all slots.
Added by the reformulation (Sections 6–7)	
$\lambda^g_{\omega t}$	$\in \mathbb{Z}_{\geq 0}$. Trucks of class (g,t) dispatched carrying column ω.
π_{nt}, π_{rt}, μ_{gt}	Duals of (55)–(56), respectively of (57) and of every further row a dispatch variable of (g,t) occupies.
n_{gt}, X	Dispatch total of class (g,t), $n_{gt} = \sum_\omega \lambda^g_{\omega t}$; the fleet $X = \sum_{g,t} n_{gt}$.

4.1 Derived quantities

Five sums recur, as *abbreviations* and not constraints.

$$\overline{V}_{nt} = \sum_{k\in K}\sum_{i\in I} V_{kint} \qquad \forall n \in N,\ t \in T \tag{1}$$

$$\overline{V^{\mathrm{e}}}_{rt} = \sum_{k\in K} V^{\mathrm{e}}_{krt} \qquad \forall r \in R,\ t \in T \tag{2}$$

$$\overline{V}_{knt} = \sum_{i\in I} V_{kint} \qquad \forall k \in K,\ n \in N,\ t \in T \tag{3}$$

$$\widehat{X}_{gikt} = \sum_{h\in H} X_{gihkt} \qquad \forall g \in G,\ i \in I,\ k \in K,\ t \in T \tag{4}$$

$$\overline{L}_{ghktr} = \sum_{i\in I} L_{gihktr} \qquad \forall g \in G,\ h \in H,\ k \in K,\ t \in T,\ r \in R(h) \tag{5}$$

The first two carry the method. The inventory block reads $\overline{V}_{nt}$ and $\overline{V^{\mathrm{e}}}_{rt}$ and nothing else about the trucks — not which truck carried what, nor how the load was placed.

4.2 Objective

Transport cost counts *dispatches*, so a configured but undispatched truck costs nothing; holding cost falls on erected RTIs.

$$\min\ z^K = \sum_{g\in G}\sum_{k\in K}\sum_{t\in T} cK\cdot Y_{gkt}, \qquad \min\ z^I = \sum_{g\in G}\sum_{n\in N}\sum_{t\in T} ic_{gn}\cdot I_{gnt} \tag{6}$$

$$\min\ \eta^K z^K + \eta^I z^I, \qquad \eta^K = \eta^I = 1 \text{ throughout,} \tag{7}$$

4.3 Inventory, flow balance and coverage

Arrivals are delayed; inventory itself accumulates one period at a time. For the plant,

$$I_{pnt} = I_{p,n,t-1} - d_{pnt} + \overline{V}_{n,t-lt} \qquad \forall n \in N,\ t \geq lt+1 \tag{8}$$

$$I_{pnt} = I_{p,n,t-1} - d_{pnt} + a_{nt} \qquad \forall n \in N,\ t \in \{2,\dots,lt\} \tag{9}$$

$$I_{pnt} = i_{pn} - d_{pnt} + a_{nt} \qquad \forall n \in N,\ t = 1 \tag{10}$$

$$Ie_{ptr} = Ie_{p,t-1,r} + \sum_{n\in M(r)} d_{pnt} - \overline{V^{\mathrm{e}}}_{rt} \qquad \forall r \in R,\ t \geq 2 \tag{11}$$

$$Ie_{ptr} = ie_{pr} + \sum_{n\in M(r)} d_{pnt} - \overline{V^{\mathrm{e}}}_{rt} \qquad \forall r \in R,\ t = 1 \tag{12}$$

Consuming a full RTI produces a folded one in the same period, which is why d_{pnt} drives (11). For the supplier,

$$I_{snt} = I_{s,n,t-1} + d_{snt} - \overline{V}_{nt} \qquad \forall n \in N,\ t \geq 2 \tag{13}$$

$$I_{snt} = i_{sn} + d_{snt} - \overline{V}_{nt} \qquad \forall n \in N,\ t = 1 \tag{14}$$

$$Ie_{str} = Ie_{s,t-1,r} + \overline{V^{\mathrm{e}}}_{r,t-lt} - \sum_{n\in M(r)} d_{snt} \qquad \forall r \in R,\ t \geq lt+1 \tag{15}$$

$$Ie_{str} = Ie_{s,t-1,r} + ae_{rt} - \sum_{n\in M(r)} d_{snt} \qquad \forall r \in R,\ t \in \{2,\dots,lt\} \tag{16}$$

$$Ie_{str} = ie_{sr} + ae_{rt} - \sum_{n\in M(r)} d_{snt} \qquad \forall r \in R,\ t = 1 \tag{17}$$

The lead time therefore acts on one side of a balance only — the sending node's stock falls in the dispatch period, the receiving node's rises lt periods later — an alignment Section 8 uses twice. Coverage stock is enforced at both nodes and on both kinds of stock,

$$I_{gnt} \ \geq\ ss_{gnt} \qquad \forall g \in G,\ n \in N,\ t \in T \tag{18}$$

$$Ie_{gtr} \ \geq\ sse_{gtr} \qquad \forall g \in G,\ r \in R,\ t \in T \tag{19}$$

the requirement being that node's own series over the next SC periods:

$$ss_{gnt} = \sum_{j=1}^{SC} d_{g,n,\tau(t,j)}, \qquad sse_{gtr} = \sum_{n \in M(r)} ss_{gnt}, \qquad \tau(t,j) = \begin{cases} t+j, & t+j \le |T|, \\ |T| - SC + j, & \text{otherwise.} \end{cases} \tag{20}$$

Near the end of the horizon (20) wraps the window back onto the last SC periods rather than truncating it, following (Díaz-Madroñero, Mula, and Peidro 2014): truncating lowers the requirement of the final periods and with it the optimal value, so the tail rule belongs to the problem statement.

4.4 The layout of a truck

Everything from here to (43) is written for one truck: one direction g, one index k, one period t. At most one stack type occupies a slot and only a configured truck has slots; a stack is oriented along one axis or the other; its height cannot exceed the trailer's, which with $r_{gr} > 0$ also empties an unused slot:

$$\widehat{X}_{gikt} \;\le\; Z_{gkt} \qquad \forall i \in I \tag{21}$$

$$LX_{gihkt} + LY_{gihkt} \;=\; X_{gihkt} \qquad \forall h \in H,\ i \in I \tag{22}$$

$$\sum_{r \in R(h)} r_{gr}\, L_{gihktr} \;\le\; rK \cdot X_{gihkt} \qquad \forall h \in H,\ i \in I \tag{23}$$

Slots must not overlap — for each pair $i < j$, the usual four-way disjunction closed by a cover row — and every slot lies inside the floor:

$$PX_{gikt} + \sum_{h \in H} \big(p_h LX_{gihkt} + q_h LY_{gihkt}\big) \le PX_{gjkt} + M^x(1 - A_{gijkt}) \tag{24}$$

$$PX_{gjkt} + \sum_{h \in H} \big(p_h LX_{gjhkt} + q_h LY_{gjhkt}\big) \le PX_{gikt} + M^x(1 - B_{gijkt}) \tag{25}$$

$$PY_{gikt} + \sum_{h \in H} \big(q_h LX_{gihkt} + p_h LY_{gihkt}\big) \le PY_{gjkt} + M^y(1 - C_{gijkt}) \tag{26}$$

$$PY_{gjkt} + \sum_{h \in H} \big(q_h LX_{gjhkt} + p_h LY_{gjhkt}\big) \le PY_{gikt} + M^y(1 - D_{gijkt}) \tag{27}$$

$$A_{gijkt} + B_{gijkt} + C_{gijkt} + D_{gijkt} \;\ge\; \widehat{X}_{gikt} + \widehat{X}_{gjkt} - 1 \tag{28}$$

$$PX_{gikt} + \sum_{h \in H} \big(p_h LX_{gihkt} + q_h LY_{gihkt}\big) \le pK \tag{29}$$

$$PY_{gikt} + \sum_{h \in H} \big(q_h LX_{gihkt} + p_h LY_{gihkt}\big) \le qK \tag{30}$$

The containment rows are unconditional, which is what lets the non-overlap rows take the tightened $M^x = pK$, $M^y = qK$.

4.5 Full truckload on every dispatched truck

A dispatched truck must be full, and it can be full in two ways: it may run out of space or of payload. A volume-only floor is not a conservative simplification but an infeasible one — filling the fraction $\alpha = 0.55$ of a 97.92 m^3 trailer ($13\,600 \times 2\,400 \times 3\,000$ mm) with freight at 579 kg/m^3 weighs 31.2 t against a 24 t cap, and seven of the industrial catalogue's fifteen types are denser still. Summing the two ratios is the opposite error: it accepts a truck 35% full by volume and 20% by weight. We therefore require a fraction α of *either* resource, a binary θ_{gkt} selecting which one binds, on the load the truck carries with each commodity's own weight:

$$\sum_{n \in N} \gamma_{p\rho(n)}\, \overline{V}_{knt} \;\ge\; \alpha\, \Gamma_K \big(Y_{pkt} - 1 + \theta_{pkt}\big) \qquad \forall k \in K,\ t \in T \tag{31}$$

$$\sum_{n \in N} w_n\, \overline{V}_{knt} \;\ge\; \alpha\, pK^w \big(Y_{pkt} - \theta_{pkt}\big) \qquad \forall k \in K,\ t \in T \tag{32}$$

$$\sum_{r \in R} \gamma_{sr}\, V^{\mathrm{e}}_{krt} \;\ge\; \alpha\, \Gamma_K \big(Y_{skt} - 1 + \theta_{skt}\big) \qquad \forall k \in K,\ t \in T \tag{33}$$

$$\sum_{r \in R} w^{\mathrm{f}}_r\, V^{\mathrm{e}}_{krt} \;\ge\; \alpha\, pK^w \big(Y_{skt} - \theta_{skt}\big) \qquad \forall k \in K,\ t \in T \tag{34}$$

Exactly one of each pair is active at $Y_{gkt} = 1$, so each is the disjunction "full by volume *or* full by payload". The same disjunction on the layer *capacity* the stacks provide rather than on the load they receive,

$$\sum_{h\in H}\sum_{r\in R(h)} \gamma_{gr}\, n_{hr}\, \overline{L}_{ghktr} \;\geq\; \alpha\,\Gamma_K\big(Y_{gkt} - 1 + \theta_{gkt}\big), \quad \sum_{h\in H}\sum_{r\in R(h)} \hat{w}_{gr}\, n_{hr}\, \overline{L}_{ghktr} \;\geq\; \alpha\, pK^{w}\big(Y_{gkt} - \theta_{gkt}\big) \tag{35}$$

is a valid tightening, kept because it lets a solver establish a layout before choosing a product assignment. Which form is the *requirement* matters. (31)–(34) is, full truckload being a property of the freight that moves; (35) follows from it because a slot never receives more RTIs of type r than its layers hold and $\hat{w}_{gr} \geq w_n$ for every $n \in M(r)$, and taken *alone* would admit trucks that are not full. That inequality is also why $\hat{w}_{pr} = \max_{n\in M(r)} w_n$ is the family *maximum*: the family *minimum* would assert that *every* product assignment of this layout fills the truck, cutting off feasible loads.

4.6 Dispatch, loading and weight

A dispatched truck must be configured,

$$Y_{gkt} \;\leq\; Z_{gkt} \qquad \forall g \in G,\ k \in K,\ t \in T, \tag{36}$$

while the converse fails: $Z_{gkt} = 1$ with $Y_{gkt} = 0$ is a truck carrying a layout and no freight, which in the reformulated block is exactly the empty column (Section 7).

Full RTIs in a slot cannot exceed the layer capacity of the stack placed there, products sharing an RTI type partitioning it; folded RTIs are capped by the truck's aggregate capacity, folded weight being deterministic in the type. Both loads are gated on the dispatch, at their own analytic bounds rather than a generic big-M:

$$\sum_{n\in M(r)} V_{kint} \;\leq\; \sum_{h\,:\,r\in R(h)} n_{hr}\, L_{p,i,h,k,t,r} \qquad \forall k \in K,\ i \in I,\ r \in R,\ t \in T \tag{37}$$

$$V^{\mathrm{e}}_{krt} \;\leq\; \sum_{h\,:\,r\in R(h)} n_{hr}\, \overline{L}_{s,h,k,t,r} \qquad \forall k \in K,\ r \in R,\ t \in T \tag{38}$$

$$V_{kint} \leq \bar{u}^{V}_{n}\, Y_{pkt} \qquad \forall k \in K,\ i \in I,\ n \in N,\ t \in T \tag{39}$$

$$V^{\mathrm{e}}_{krt} \leq \bar{u}^{\mathrm{e}}_{r}\, Y_{skt} \qquad \forall k \in K,\ r \in R,\ t \in T \tag{40}$$

The payload is a property of the truck, the crush limit of each stack; on plant-bound trucks that cap is *period-exact*, evaluated on the mix actually loaded in the slot, on supplier-bound trucks deterministic in the layout:

$$\sum_{n\in N} w_n\, \overline{V}_{knt} \;\leq\; pK^{w}\, Y_{pkt}, \qquad \sum_{r\in R} w^{\mathrm{f}}_{r}\, V^{\mathrm{e}}_{krt} \;\leq\; pK^{w}\, Y_{skt} \qquad \forall k \in K,\ t \in T \tag{41}$$

$$\sum_{n\in N} w_n\, V_{kint} \;\leq\; W^{\max} \qquad \forall k \in K,\ i \in I,\ t \in T \tag{42}$$

$$\sum_{h\in H}\sum_{r\in R(h)} w^{\mathrm{f}}_{r}\, n_{hr}\, L_{s,i,h,k,t,r} \;\leq\; W^{\max} \qquad \forall i \in I,\ k \in K,\ t \in T \tag{43}$$

where (41) names both payload rows and (39) is tighter in the relaxation than their implication, since what bounds a slot's content is the crush cap.

4.7 Symmetry, slot-set sizing and data assumptions

The slots of a truck are interchangeable, and so are the trucks of a direction and period. Both are ordered:

$$\widehat{X}_{g,i+1,k,t} \;\leq\; \widehat{X}_{gikt} \qquad \forall g \in G,\ k \in K,\ t \in T,\ i < |I| \tag{44}$$

$$Y_{g,k+1,t} \;\leq\; Y_{gkt} \qquad \forall g \in G,\ t \in T,\ k < |K| \tag{45}$$

Re-indexing makes (45) valid, but it removes only *which* subset of trucks is used: loads and layouts stay interchangeable among those used, so a plan dispatching m trucks in a period still appears $m!$ times — the difficulty Section 5 removes by aggregation rather than by a row. The slot count is a modelling parameter, and

$$|I| \;\leq\; \Big\lfloor (pK \cdot qK) \big/ \min_{h\in H} p_h q_h \Big\rfloor \tag{46}$$

eliminates provably unused slot variables without cutting any feasible layout. It is more than preprocessing: a placement model given fewer slots than a layout needs can report infeasible while the load is placeable (Section 8).

Four properties of the data are assumed: ρ is a partition; $lt \geq 1$, which the lagged terms require; all weights and heights are strictly positive, which makes $\bar{u}_n^V$ and $\bar{u}_r^{\mathrm{e}}$ well defined and lets (23) empty an unused slot; and the footprint geometry is consistent, one layer of stack type h tiling its footprint, $n_{hr}a_r = p_h q_h$, with the volume charged to an RTI its bounding box, $\gamma_{gr} = a_r r_{gr}$. *The last is a departure*, and Proposition 2 and the floor-area inequality (72) are both false without it. It costs nothing here, both instance families building one stack type per distinct RTI footprint, so $n_{hr} = 1$ and the relation is an identity.

4.8 Domains

$$Z_{gkt},\, Y_{gkt},\, \theta_{gkt},\, X_{gihkt},\, LX_{gihkt},\, LY_{gihkt},\, A_{gijkt},\, B_{gijkt},\, C_{gijkt},\, D_{gijkt} \in \{0,1\},$$
$$L_{gihktr},\, V_{kint},\, V^{\mathrm{e}}_{krt},\, I_{gnt},\, Ie_{gtr} \in \mathbb{Z}_{\geq 0}, \qquad PX_{gikt},\, PY_{gikt} \geq 0.$$

5 Why branch and price, and why this decomposition

Two structural difficulties stand between the formulation of Section 4 and an industrial instance. The model carries a complete two-dimensional packing problem for every truck of every direction and period — $2|K||T|$ replicas, each contributing $4\binom{|I|}{2}$ big-M disjunctions (24)–(27) whose linear relaxation is weak — and those replicas are interchangeable, any two used trucks of one direction and period being able to swap layouts and loads, so every solution appears a factorial number of times. The second difficulty scales worst and does not depend on the loading geometry at all, being present in the volumetric benchmark of Appendix A exactly as in 3D-SCPTOP.

The Dantzig–Wolfe reformulation below (Dantzig and Wolfe 1960; Barnhart et al. 1998; Desaulniers, Desrosiers, and Solomon 2005) removes both, and removes the first without restricting anything. Figure 1 is the picture, Table 3 the row-by-row detail.

Blocks. Take the block index to be $\mathcal{B} = G \times K \times T$. Every row from (21) to (45) — the first column of Table 3 names them — carries a single direction, truck and period as a fixed $\forall$ index or a literal, and none couples two values of g, k or t; the exception is (45), relating truck $k+1$ to truck k. The remaining rows are the objective, the inventory and coverage block (8)–(19) and the two definitions (1)–(2), and they see the blocks *only* through those aggregates and the count of dispatches z^K charges for. Block-angularity over $\mathcal{B}$ is therefore a property of the formulation, not a discovery about it.

Delete the quantifier. Fix a class (g,t), take the block rows, and *remove the* $\forall$ *over* $\mathcal{B}$: drop the indices g, k and t from every variable and read g as data. The stack-height row, for instance, goes

$$\underbrace{\sum_{r \in R(h)} r_{gr} L_{gihktr} \leq rK\, X_{gihkt} \quad \forall g,h,i,k,t}_{\text{3D-SCPTOP, (23)}} \longmapsto \underbrace{\sum_{r \in R(h)} r_{gr} L_{ihr} \leq rK\, X_{ih} \quad \forall h,i,}_{(\mathrm{PP}^g),\ (63)} \tag{47}$$

and every other block row the same way; what is left is a mixed integer programme over one truck, the (PP^g) of Section 7. Nothing is invented for the subproblem and nothing of the geometry is left behind in the master. Two rows do not make the journey: (45), which the aggregation disposes of, and (36), whose two binaries collapse into one (Section 7).

Identical blocks. K is anonymous, so the blocks of a fixed (g,t) are identical and are aggregated: in place of one copy per truck, the reformulation carries one non-negative integer variable per block *solution*, counting how many of the class's blocks take it, and one row per class bounding that count by $|K|$. This is where the second difficulty disappears — the trucks are never distinguished, so there is no permutation to search — and it is why the dropped row costs nothing: counts carry no order, so the master has no variable in which to express (45). The converse direction of Proposition 1 assigns trucks *in index order*, so the equivalence holds against the compact model *with* that row in force.

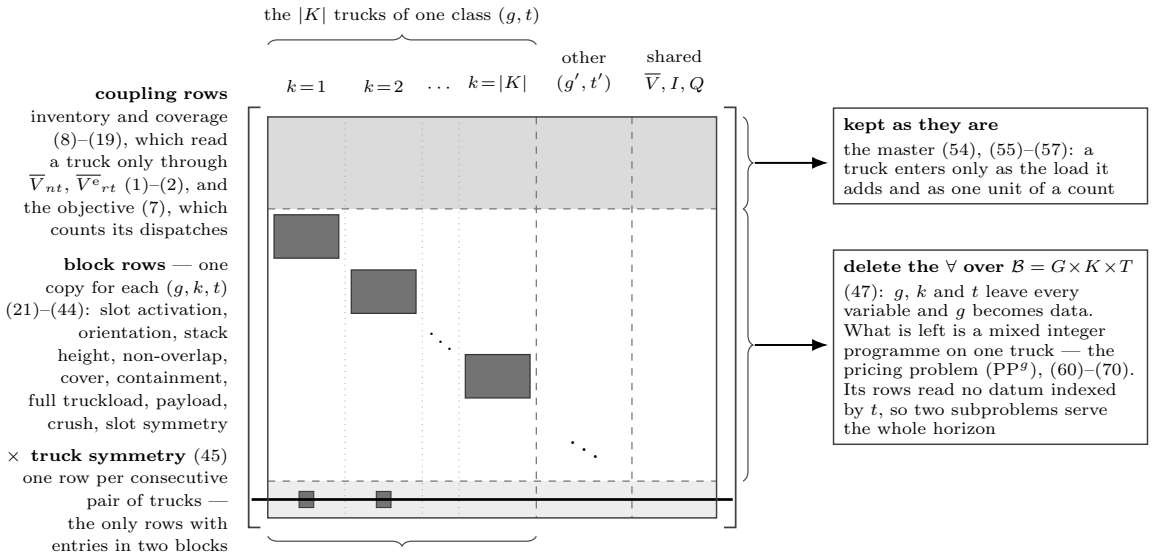


Figure 1: The decomposition is read off the formulation, not chosen by analogy. Every geometric, stacking, crush, payload and full-truckload row of Section 4 is confined to one truck, so the matrix is block-angular over $\mathcal{B} = G \times K \times T$; deleting the quantifier over $\mathcal{B}$ from those rows leaves a mixed integer programme on a single truck, and that programme is the pricing problem. Nothing is invented for the subproblem and no geometry stays behind in the master, which is why a branching row added to the master never changes what the subproblem is. Table 3 is the same statement row by row.

Classes. The classes are the pairs (g,t), and within a class the block rows are pinned to the class's g, so there is one pricing problem per *direction*. The pricing rows read no datum indexed by t — the trailer, the stack types, the crush limit and the floor are time-invariant — so the periods of one direction differ *only in their duals*, one column set serves them all, and the whole horizon costs two pricing problems rather than $2|T|$.

Discretisation. Every block variable that reaches the master is integer, so a block solution may be taken to be an *integer* point of its pricing problem completed by placement coordinates. The multipliers are then integer rather than merely non-negative, and a fractional one is a genuine integrality violation.

Table 3: The block rows of Section 4 and the rows of (PP^g) they become when the $\forall$ over $\mathcal{B} = G \times K \times T$ is deleted. The two directions share the geometry, so it is written once with g as data; the direction-specific rows are those of Section 7.1. One row of (PP^g) is not a transcription and is marked as such.

Row	Section 4	(PP^g)
Block rows: the pricing problem		
Slot active	(21)	(61)
Orientation	(22)	(62)
Stack height	(23)	(63)
Non-overlap	(24)–(27)	(64)–(65)
Cover	(28)	(66)
Containment	(29)–(30)	(67)
Full truckload, volume	(31), (33)	(69)
Full truckload, payload	(32), (34)	(70)

Row	Section 4	(PP^g)
The same on capacity	(35)	(71), re-gated on Z
Occupied floor area	— (no counterpart)	(72), a valid inequality
Slot capacity	(37)	(73)
Folded capacity	(38)	(77)
Dispatch gate, full	(39)	(74)
Dispatch gate, folded	(40)	(78)
Payload, both directions	(41)	(75), (79)
Crush limit, plant-bound	(42)	(76)
Crush limit, supplier-bound	(43)	(80)
Slot symmetry	(44)	(68)
Transport cost	(6)	$\eta^K cK \cdot Z$ in (60)
Dispatch	(36)	deleted; Z carries the truck
Truck symmetry	(45)	dropped in the aggregation
Coupling rows: the master		
Aggregate flows	(1)–(2)	(55)–(56)
Fleet per class	$\lvert K\rvert$ trucks per (g,t)	(57)
Inventory and coverage	(8)–(19)	verbatim in the master
Holding cost	(6)	second term of (54)

6 The master problem

6.1 Columns

A column of a direction is the image of a feasible point of that direction's pricing problem, together with the point itself as a witness; Ω^p and Ω^s are the two column sets, indexed by ω. A *plant-bound* column $\omega \in \Omega^p$ is a vector $b_\omega = (b_{\omega n})_{n\in N}$ of full RTIs — the value of $\overline{V}_n$ at the point — for which there exist a slot count $m \le |I|$, stack types $h_1, \dots, h_m \in H$, orientations, bottom-left coordinates (x_i, y_i), layer counts ℓ_{ir} and per-slot product loads v_{in} such that

$$\text{the } m \text{ footprints, rotated as chosen, do not overlap and lie inside } pK \times qK, \tag{48}$$

$$\textstyle\sum_{r\in R(h_i)} r_{pr}\ell_{ir} \;\le\; rK \qquad \forall i \le m, \tag{49}$$

$$\textstyle\sum_{n\in M(r)} v_{in} \;\le\; n_{h_i r}\ell_{ir} \qquad \forall i \le m,\ r \in R, \tag{50}$$

$$\textstyle\sum_{n\in N} w_n v_{in} \;\le\; W^{\max} \qquad \forall i \le m, \tag{51}$$

$$\textstyle\sum_{n\in N} w_n b_{\omega n} \;\le\; pK^w, \qquad \max\Big\{\frac{\sum_n \gamma_{p\rho(n)} b_{\omega n}}{\Gamma_K},\ \frac{\sum_n w_n b_{\omega n}}{pK^w}\Big\} \;\ge\; \alpha, \tag{52}$$

$$b_{\omega n} = \textstyle\sum_{i\le m} v_{in} \qquad \forall n \in N. \tag{53}$$

These are the block rows of Section 4 read on one truck (Table 3); in (50) $n_{h_i r}\ell_{ir}$ is read as 0 for $r \notin R(h_i)$, and it is an *inequality* because a partly filled layer is a legal compact layout.

A *supplier-bound* column is the analogous object $e_\omega = (e_{\omega r})_{r\in R}$ over folded heights, weights and volumes, with (43) as its per-slot crush condition and $e_{\omega r} \le \sum_{i\le m} n_{h_i r}\ell_{ir}$, which is (38), in place of (50) and (53); the weight of a folded RTI depends only on its type, so no product assignment is needed. We write (48)–(53) from here on for either direction, writing $\mathrm{image}_c(\omega)$ for the entry of b_ω or e_ω at the *commodity* c — a product when $g = \mathrm{p}$, an RTI type when $g = \mathrm{s}$. Every column costs $c_\omega = \eta^K cK$, and Ω^g holds the *non-empty* columns only, the all-zero point being the slack of (57).

Two features of this definition carry the method: the column holds the per-slot assignment v_{in}, so (51) is the period-exact cap (42) and not the family surrogate a catalogue of capacity-only configurations must use, and it retains its witnessing placement, so a plan is a set of drawings.

6.2 The master

Let $\lambda^g_{\omega t} \in \mathbb{Z}_{\ge 0}$ be the number of trucks of class (g,t) dispatched carrying column ω. The master problem (MP) is

$$\min \sum_{g\in G}\sum_{t\in T}\sum_{\omega\in\Omega^g} c_\omega\, \lambda^g_{\omega t} \;+\; \eta^I \sum_{g\in G}\sum_{n\in N}\sum_{t\in T} i c_{gn} I_{gnt} \tag{54}$$

subject to

$$\overline{V}_{nt} = \sum_{\omega \in \Omega^p} b_{\omega n}\, \lambda^p_{\omega t} \qquad \forall n \in N,\ t \in T \qquad (55)$$

$$\overline{V^{\mathrm{e}}}_{rt} = \sum_{\omega \in \Omega^s} e_{\omega r}\, \lambda^s_{\omega t} \qquad \forall r \in R,\ t \in T \qquad (56)$$

$$\sum_{\omega \in \Omega^g} \lambda^g_{\omega t} \le |K| \qquad \forall g \in G,\ t \in T \qquad (57)$$

(8)–(19) verbatim

$\lambda^g_{\omega t} \in \mathbb{Z}_{\ge 0}$, I_{gnt}, $Ie_{gtr} \in \mathbb{Z}_{\ge 0}$, $\overline{V}_{nt}$, $\overline{V^{\mathrm{e}}}_{rt} \in \mathbb{Z}_{\ge 0}$ (implied by (55)–(56)).

That is the whole master: every column costs the same, so the transport term is cK times the number dispatched, and (55)–(56) are (1)–(2) with each truck's load replaced by its column's image. Since Ω^g holds the non-empty columns only, $\overline{\lambda}_{gt} = \sum_\omega \lambda^g_{\omega t}$ is the number *dispatched* by class (g, t), which is what Section 8 branches on.

Proposition 1 (equivalence)**.** *MP and 3D-SCPTOP have the same optimal value.*

Proof. Each used truck (g, k, t) of a 3D-SCPTOP solution induces a load vector satisfying its direction's column conditions, read off its own layout with $\ell_{ir} = L_{g,i,h_i,k,t,r}$ and, at the plant, $v_{in} = V_{kint}$; every one of those conditions is a row the truck already satisfies. Counting trucks by load vector gives a feasible MP solution of equal cost, since transport cost depends only on the number of used trucks and the inventory rows see only the aggregates (55)–(56) reproduce. Conversely, give each truck of an MP solution the layout witnessing its column: the blocks are identical, so any assignment serves, and in index order the result also satisfies (45). □

No hypothesis is needed because there is no catalogue of layouts to be sized: in Section 4 a truck's layout is its own. MP is therefore the same problem written differently, and any bound derived from it bounds the original.

6.3 The restricted master and the reduced cost

$|\Omega^g|$ is astronomical, so MP is solved over a subset $\bar{\Omega}^g$ with integrality relaxed — the restricted master problem (RMP) — whose duals the pricing problems use to look for columns of negative reduced cost (Figure 2). With π_{nt}, π_{rt} the duals of (55)–(56) and $\mu_{gt} \le 0$ that of (57), the reduced cost of a plant-bound column dispatched in t is

$$\bar{c}(\omega, t) \;=\; c_\omega - \sum_{n \in N} \pi_{nt}\, b_{\omega n} - \mu_{pt}, \qquad (58)$$

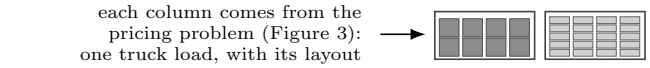
and analogously at the supplier. The pricing problem of class (g, t) minimises $c(x) - \sum_c \pi_{ct}\, \mathrm{image}_c(x)$ over the feasible points of the direction's block, so with optimal value $z^\star_{gt}$ it gives $\min_\omega \bar{c}(\omega, t) = z^\star_{gt} - \mu_{gt}$, and a column enters when that is negative.

The prices, irrational in general, must be rounded *upwards* before entering an integer-coefficient subproblem, never to nearest, so that the rounded objective dominates the true one and the subproblem's value stays an over-estimate of the true maximum, as the certificate below requires. Rounding to nearest invalidates that certificate and, with reduced costs near zero, returns a value just above the threshold so that convergence is never declared.

The RMP is *elastic*: non-negative artificials sit on (55)–(56), on the coverage rows and on the cuts below at a large price, so it is feasible for any pool and always yields duals, and a solution with positive artificials is not a plan. The elasticity is one-sided where it matters: a definition row reads $\overline{V} = \sum_\omega b\lambda + \mathrm{art}$, $\mathrm{art} \ge 0$, so slack can only enlarge a delivery — which is why stock non-negativity stays rigid against a fleet lower bound.

It also carries cumulative cover cuts. With ρ^g_{ct} the cumulative requirement of commodity c at the receiving node g through t — demand to date plus the coverage stock of t, net of initial stock and freight in transit, floored at zero — and $\bar{u}^g = \min\{\lfloor pK^w / \min_c w_c \rfloor, \lfloor \Gamma_K / \min_c \gamma_{gc} \rfloor\}$ an upper bound on the units one truck holds, every $t > lt$ gives

$$\sum_{\tau \le t - lt} \sum_{\omega \in \Omega^g} \lambda^g_{\omega\tau} \;\ge\; \max\Big\{ \Big\lceil \tfrac{\sum_c \gamma_{gc} \rho^g_{ct}}{\Gamma_K} \Big\rceil,\ \Big\lceil \tfrac{\sum_c w_c \rho^g_{ct}}{pK^w} \Big\rceil,\ \Big\lceil \tfrac{\sum_c \rho^g_{ct}}{\bar{u}^g} \Big\rceil \Big\}. \qquad (59)$$



	$\lambda^{p}_{\omega t}$ full RTIs	$\lambda^{s}_{\omega' t}$ folded RTIs	$\overline{V}_{nt}$, $\overline{V^{e}}_{rt}$ aggregate	I_{gnt}, Ie_{gtr} stock	artificial columns† priced M	to the pricing problem, rounded up
objective (54)	cK	cK	·	ic_{gn}	M	
aggregate flow (55) one row per product n, period t	$b_{\omega n}$	·	-1	·	$+1$	π_{nt}
aggregate flow (56) one row per RTI type r, period t	·	$e_{\omega' r}$	-1	·	$+1$	π_{rt}
truck count (57) one row per direction and period	1	1	·	·	·	μ_{gt}
cumulative cover cut (59)† one row per direction and period, over the dispatches with $\tau \le t - lt$	1	1	·	·	$+1$	folds into μ_{gt}
inventory balance (8)–(17) the receiver's arrival lagged by lt	·	·	± 1	± 1	·	no λ: pricing feels these rows only through π
coverage stock (18)–(19)	·	·	·	$+1$	$+1$	
rows a catalogue of loading configurations would add	**none.** No slot, stack type, orientation, coordinate or layer count is a variable of the master: the geometry stays inside the column, with its witnessing layout (Figure 3).					

† Rows and columns of the *restricted* master; (54)–(57) together with (8)–(19) are the master itself.

Figure 2: The master is a flow model that has never seen a box. A column is one truck load, and the only rows it enters are the two aggregate-flow definitions, one truck count per direction and period, and the cumulative cover cut — every one of which merely counts or sums what the load contains. No slot, stack type, orientation, coordinate or layer count is a variable here, so there is no catalogue of loading configurations to enumerate, and no row written over these variables can reach the geometry; the inventory and coverage rows see the trucks only through $\overline{V}$ and $\overline{V^{e}}$, and all that ever reaches the pricing problem is π and μ. The non-negative artificials keep the restricted master feasible, and its duals defined, for any pool — a solution that uses them is not a plan.

Each ratio divides a requirement by the most one truck can carry of that resource, so over-estimating $\bar{u}^{g}$ only weakens the cut while the shift by lt is mandatory. Its duals fold into μ_{gt} and leave (58) exact, which is possible exactly when a row $\sum_{\omega} \phi(\omega)\lambda_{\omega} \lessgtr \beta$ has ϕ constant within a class or affine in the column's image. Such a row is *robust*: it changes the duals and nothing else. Every row this paper adds is of that kind, which is why no branching decision reaches a pricing problem.

Finally the RMP is *period-scoped*: a column priced at t receives a dispatch variable at t alone. That changes the restricted master's size and not the master's, so a sweep that finds nothing anywhere still proves the restricted LP equal to the master LP.

6.4 What may be reported as a lower bound

The natural candidate is not a bound at all. The RMP is a *restriction* of MP, so $z_{\mathrm{RMP}} \ge z_{\mathrm{MP}}$: until column generation converges the RMP value is an *upper* bound on the master LP. Its one legitimate use is as a filter.

A certificate needs a full sweep. When the pricing problem has *proved* that

$$\zeta_{gt} \;:=\; \max_{\omega \in \bar{\Omega}^{g}} \sum_{c} \pi_{ct}\, \mathrm{image}_{c}(\omega) \;\le\; \eta^{K} cK - \mu_{gt} \qquad \forall g \in G,\ t \in T,$$

nothing outside $\bar{\Omega}^{g}$ prices out, the duals are feasible for the dual of the full MP, and weak duality gives $z_{\mathrm{RMP}} = z_{\mathrm{MP}} \le z^{\star}_{\text{3D-SCPTOP}}$: a bound as good as the sweep behind it (Algorithm 1). Memoising duals, and

pricing only the pairs whose aggregates are fractional, are legitimate while columns are still arriving and forfeit the right to certify; so does one pair stopped on its time limit. One case looks like a time limit and is not: a call stopped by a cutoff *without* returning a point ran out of search tree rather than out of time and does prove that no column beats the threshold, the two being separated only by whether a point came back. Every branching fix must be released before the sweep, a restricted LP sitting above the region's own. The bound is then the *raw* LP value, penalty included; positive artificials at convergence mean the region admits no plan at all and it is pruned as infeasible.

Algorithm 1 CERTIFY($\bar{\Omega}, \mathcal{R}$) — the sweep a lower bound rests on. No memoisation, no restriction to the pairs a branch has touched, every branching fix released.

release every branching fix; impose $\mathcal{R}$ on the RMP as fleet rows
loop
  solve the LP over $\bar{\Omega}$ inside $\mathcal{R}$
  if infeasible **then**
    return $(\infty,\ \text{true})$ ▷ region refuted
  end if
  proved $\leftarrow$ true
  for all $(g,t) \in G \times T$ **do**
    $\zeta_{gt} \leftarrow$ PRICE(π, μ, g, t) to proven optimality, duals rounded *up*
    if the call returned neither OPTIMAL nor INFEASIBLE, and was not an objective-limit stop with *no* point returned **then**
      proved $\leftarrow$ false ▷ a cutoff that returns nothing is still a proof
    end if
  end for
  if some ζ_{gt} prices out **then**
    add its columns to $\bar{\Omega}$ and **continue**
  end if
  if artificials > 0 **then**
    return $(\infty,\ \text{true})$ ▷ no plan over Ω^g lies in $\mathcal{R}$
  end if
  return (raw LP value, proved) ▷ a bound only if proved
end loop

A bound without convergence. A complete sweep that finds columns still yields a Lagrangian bound, $\bar{z} + \sum_{g,t} |K| \min(0, \bar{c}^\star_{gt})$ with the count row (57) as multiplier, provided each $\bar{c}^\star_{gt}$ bounds the pricing *maximum* from above — for which dropping the geometry, the integrality and the floor leaves a two-resource knapsack. It is too weak to rely on here: a reduced cost of $-\varepsilon$ at every pair subtracts $2|T||K|\varepsilon$ from $\bar{z}$, where the distance to make up is a fraction of one truck.

A bound by inclusion. Every 3D lower bound in this paper is an instance of the following, which is affordable where the certificate above — with the geometric pricing problem as oracle — is not.

Proposition 2 (bound by inclusion)**.** *Let $\Omega^g_{\text{vol}} \supseteq \Omega^g$ be the loads that fit the trailer's volume and the payload and clear the full-truckload floor on the binding resource, and z_{vol} the certified LP value of the master* (54)–(57) *over them. Then $z_{\text{vol}} \le z^\star_{\text{3D-SCPTOP}}$, and the same holds inside any region defined by bounds on the fleet vector.*

Proof. The inclusion rests on two conventions of Section 4: $\Gamma_K = pK \cdot qK \cdot rK$, and the volume γ_{gr} charged to an RTI is its bounding box, with no allowance for void space inside it or clearance between stacks, so that by footprint consistency one layer of stack type h occupies exactly $n_{hr}\gamma_{gr} = p_h q_h r_{gr}$. A placement satisfying (48)–(53) puts its RTIs in disjoint boxes inside the trailer — footprints non-overlapping and inside $pK \times qK$, each stack under rK by (49) — so their bounding-box volumes sum to at most Γ_K, its weight is at most pK^w, and (52) is the same floor in both sets; hence $\Omega^g \subseteq \Omega^g_{\text{vol}}$. Only the column set depends on the load model, so the 3D master is the volumetric master with columns removed: its LP value is at least z_{vol}, and its optimum at least its LP value. □

The consequence is a division of labour. A geometric sweep costs $2|T|$ calls at the pricing limit; over the volumetric relaxation the same sweep is a two-constraint knapsack oracle and closes in seconds. The method therefore asks geometric pricing only for columns and takes its bound from the volumetric relaxation, so the gap between them is a genuine optimality gap for the 3D problem.

7 The pricing problem

By the Blocks and Classes steps of Section 5 the pricing problem of class (g,t) *is* the block of 3D-SCPTOP with the quantifier over $\mathcal{B}$ removed: the same rows, with g supplying the data and the master the duals, the only rows below with no counterpart in Section 4 being the valid inequalities (72). There is one such problem per direction, written (PP^g). With c the commodity index of Section 6.1, image_c, w_c, γ_{gc} and r_{gc} are $\overline{V}_n, w_n, \gamma_{p\rho(n)}, r_{p\rho(n)}$ respectively $V^{\mathrm{e}}_r, w^{\mathrm{f}}_r, \gamma_{sr}, r_{sr}$; the partition programme (82)–(86) uses the same convention. Figure 3 draws what the block designs.

$$\min\ \eta^K cK\cdot Z\ -\ \sum_c \pi_{ct}\,\text{image}_c \qquad \bar{c} = z - \mu_{gt} \tag{60}$$

$$\widehat{X}_i \le Z \qquad \forall i\in\mathcal{I} \tag{61}$$

$$LX_{ih} + LY_{ih} = X_{ih} \qquad \forall h\in H,\ i\in\mathcal{I} \tag{62}$$

$$\sum_{r\in R(h)} r_{gr}\,L_{ihr} \le rK\cdot X_{ih} \qquad \forall h\in H,\ i\in\mathcal{I} \tag{63}$$

$$PX_i + \delta^x_i \le PX_j + M^x(1-A_{ij}),\quad PX_j + \delta^x_j \le PX_i + M^x(1-B_{ij}) \qquad \forall i<j \tag{64}$$

$$PY_i + \delta^y_i \le PY_j + M^y(1-C_{ij}),\quad PY_j + \delta^y_j \le PY_i + M^y(1-D_{ij}) \qquad \forall i<j \tag{65}$$

$$A_{ij} + B_{ij} + C_{ij} + D_{ij} \ge \widehat{X}_i + \widehat{X}_j - 1 \qquad \forall i<j \tag{66}$$

$$PX_i + \delta^x_i \le pK,\qquad PY_i + \delta^y_i \le qK \qquad \forall i\in\mathcal{I} \tag{67}$$

$$\widehat{X}_{i+1} \le \widehat{X}_i \qquad \forall i \le |\mathcal{I}|-1 \tag{68}$$

$$\sum_c \gamma_{gc}\,\text{image}_c\ \ge\ \alpha\,\Gamma_K\,(Z-1+\theta) \tag{69}$$

$$\sum_c w_c\,\text{image}_c\ \ge\ \alpha\,pK^w\,(Z-\theta) \tag{70}$$

with $\delta^x_i = \sum_h (p_h LX_{ih} + q_h LY_{ih})$ and $\delta^y_i = \sum_h (q_h LX_{ih} + p_h LY_{ih})$ the footprint of slot i in its chosen orientation, and (67) naming both containment rows and (64), (65) the four non-overlap rows, so that at $Z=1$ the pair (69)–(70) is the disjunction (52).

Deleting a variable is a claim about the feasible set. With no loading-configuration index, Z_{gkt} and Y_{gkt} differ at exactly one kind of point, $Z=1$ with $Y=0$: a truck carrying a layout and no freight, whose image and cost are both 0 by (39)–(40) and (6) — precisely those of the empty column. Identifying the two binaries therefore removes no element of Ω^g, and the master, which sees only images and costs, cannot tell the difference. The block keeps Z, charges it $\eta^K cK\cdot Z$ and hangs every load gate on it; the empty column, analytically known with reduced cost $-\mu_{gt}\ge 0$, is excluded by fixing $Z=1$.

Four implied rows are available and the oracles behind Section 9 carry the second pair — two restating the floor on the layer *capacity* rather than on the load, and two bounding the occupied floor area:

$$\sum_{h,\,r\in R(h)} \gamma_{gr}\,n_{hr}\,\overline{L}_{hr}\ \ge\ \alpha\,\Gamma_K\,(Z-1+\theta),\qquad \sum_{h,\,r\in R(h)} \hat{w}_{gr}\,n_{hr}\,\overline{L}_{hr}\ \ge\ \alpha\,pK^w\,(Z-\theta), \tag{71}$$

$$\sum_i\sum_h p_h q_h X_{ih}\ \le\ pK\cdot qK,\qquad \sum_i\sum_h p_h q_h X_{ih}\ \ge\ \alpha\,pK\cdot qK\,(Z-1+\theta). \tag{72}$$

(71) is (35) with the block index dropped. (72) is a valid inequality rather than a transcription, and is where the footprint consistency of Section 4.7 earns its place: its first row says the footprints lie on the floor, and for the second, multiplying (63) by $p_h q_h$ and using $n_{hr}\gamma_{gr} = p_h q_h r_{gr}$ gives

$$\sum_{r\in R(h)} \gamma_{gr}\,n_{hr}\,L_{ihr}\ =\ p_h q_h \sum_{r\in R(h)} r_{gr}\,L_{ihr}\ \le\ p_h q_h\,rK\,X_{ih},$$

so the volume the stacks hold is at most rK times the occupied area; summing over i and h and chaining through (71) gives the row, and a truck full by volume therefore has at least a fraction α of its floor occupied. Without the tiling assumption the step fails and the row can cut off feasible layouts.

7.1 The direction-specific rows

Plant-bound, $g = \mathrm{p}$, with $\text{image}_n = \overline{V}_n$:

$$\sum_{n \in M(r)} V_{in} \;\le\; \sum_{h \,:\, r \in R(h)} n_{hr}\, L_{ihr} \qquad \forall i \in \mathcal{I},\, r \in R \tag{73}$$

$$V_{in} \;\le\; \bar{u}^{V}_{n}\, Z \qquad \forall i \in \mathcal{I},\, n \in N \tag{74}$$

$$\sum_{n \in N} w_n\, \overline{V}_n \;\le\; pK^{w}\, Z \tag{75}$$

$$\sum_{n \in N} w_n\, V_{in} \;\le\; W^{\max} \qquad \forall i \in \mathcal{I} \tag{76}$$

and supplier-bound, $g = \mathrm{s}$, with $\text{image}_r = V^{\mathrm{e}}_r$:

$$V^{\mathrm{e}}_r \;\le\; \sum_{h \,:\, r \in R(h)} n_{hr}\, \overline{L}_{hr} \qquad \forall r \in R \tag{77}$$

$$V^{\mathrm{e}}_r \;\le\; \bar{u}^{\mathrm{e}}_r\, Z \qquad \forall r \in R \tag{78}$$

$$\sum_{r \in R} w^{\mathrm{f}}_r\, V^{\mathrm{e}}_r \;\le\; pK^{w}\, Z \tag{79}$$

$$\sum_{h} \sum_{r \in R(h)} w^{\mathrm{f}}_r\, n_{hr}\, L_{ihr} \;\le\; W^{\max} \qquad \forall i \in \mathcal{I} \tag{80}$$

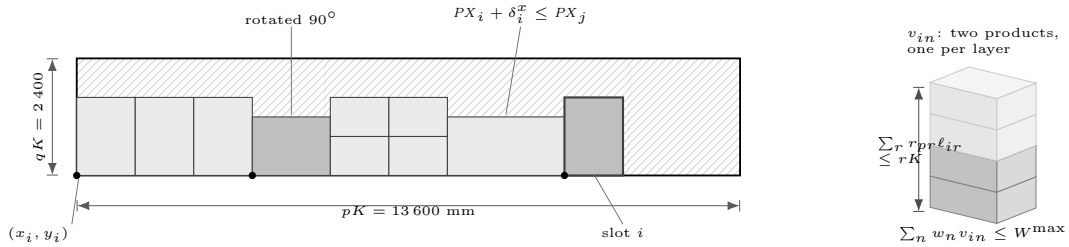
The asymmetry is the compact model's own: the crush limit on a plant-bound truck depends on which products ride in the slot, so it needs the per-slot assignment V_{in}, and a supplier-bound truck does not.

What the method requires of the oracle. The oracle's only obligation is to return an optimal point of *this* feasible set for the duals it is handed, and three conditions make its answer usable. The slot set must be the true bound (46), an optimum proved over fewer slots bounding a different problem. The floor must be imposed on the load carried, with each commodity's own weight. And the block must accept a per-commodity cap on the image, and an exact image, which is what Sections 8.1 and 8.3 ask of it: an oracle that silently ignores such a request answers a different question. The hard core is the disjunctive relation (64)–(66), whose linear relaxation is weak, and which may equivalently be imposed as one two-dimensional disjointness relation over the slot rectangles with no separating binaries at all.

The volumetric oracle. The same master is also solved over the loads of the volumetric benchmark of Appendix A, where a load is a column iff its volume fits the trailer's, its weight the payload, and it clears the floor on the binding resource. Pricing is then a two-constraint integer knapsack, which proves optimality routinely and makes a certificate affordable at every node. Running both oracles through one master makes Section 9.5 an experiment about the loading representation rather than about two solution technologies.

A pricing ladder. Any column of negative reduced cost improves the restricted master; the *most* negative one is needed only for the certificate. Geometric column generation therefore runs as a ladder whose first rung is a randomised greedy construction over enumerated stack fills, placed by a shelf packer and run *under the current duals*, keeping the best two loads per pair that beat the pair's threshold by half a percent and sending the remaining pairs to the exact oracle. That rung is pricing output and never seeding, and the certifying sweep never uses it.

What pricing cannot return. For any dual vector (60) is linear in the image, so it attains its optimum at an extreme point of $\mathrm{conv}(\Omega^g)$: pricing returns extreme loads only. A balanced load inside the hull is dominated componentwise by a fractional blend of single-commodity trucks — the relaxation buys that blend for one truck where an integer plan pays for each truck it mixes — so no choice of duals returns it, however the search moves them (Figure 4). Interior loads are precisely what an integer plan needs when a period's demand does not decompose into full trucks, which is why the method has a second generator for them.



(a) the floor, (64)–(67)

(b) one slot, (49) and (51)

Figure 3: What the pricing problem designs. A column is a drawing a loading crew could execute: footprints placed on the floor without overlap and inside $pK \times qK$, each slot a stack of homogeneous layers under the height budget (63) and the per-slot crush cap, the load assigned to the slots in the same period so that $b_{\omega n} = \sum_{i \le m} v_{in}$, and the whole truck full on the binding resource (69)–(70): volume *or* payload, $\max\{\gamma/\Gamma_K,\, w/pK^w\} \ge \alpha$, and never the sum of the two ratios.

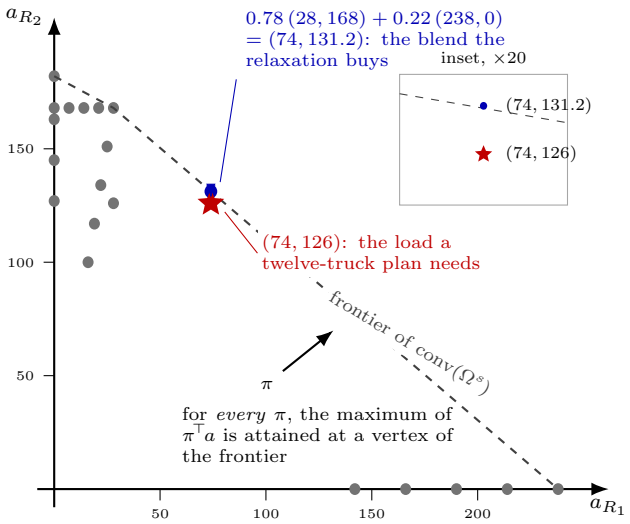


Figure 4: Why the load a cheaper plan needs is never priced. Dots are the supplier-bound loads obtained from a sweep of demand-derived price vectors; the dashed line is the upper-right frontier of their convex hull. The pricing objective is linear, so for *any* dual vector π its optimum is attained at a vertex of that frontier. The load $(74, 126)$ lies strictly below it — it is dominated componentwise by the blend $0.78\,(28, 168) + 0.22\,(238, 0) = (74, 131.2)$ — so no choice of duals returns it. The relaxation buys that blend for one truck, while an integer plan must pay for both of the trucks it mixes. The construction of Section 8.2 reaches the interior directly: it partitions a delivery vector into legal loads instead of asking a linear objective for them.

8 Branching, with the flow-first construction inside it

The branching hierarchy is read off the robustness criterion of Section 6.3. A row with $\phi \equiv 1$ — any bound on a sum of dispatch counts, including the per-class total n_{gt} and the fleet $X = \sum_{g,t} n_{gt}$ — has its dual folded into μ_{gt} exactly as (57) does, and all such aggregates are integer in every plan. A row with ϕ affine in the image, a bound on $\overline{V}_{nt}$ or $\overline{V^{\mathrm{e}}}_{rt}$, is absorbed into π_{ct}, already an input of the pricing problem; these count physical RTIs, so they too may carry a dichotomy. Bounding a single $\lambda^g_{\omega t}$ is neither: honouring it means excluding that column from the subproblem (Vanderbeck 2000), destroying the property that one column set serves every period. It is also useless here, the master holding many columns that differ in a single RTI, so the relaxation answers by shifting the same fractional weight one column over — measured at 3 000 branches with no integral relaxation reached.

Between the two robust rungs the first binds, because transport cost is $cK \cdot X$ and nothing else in the

objective counts trucks. A bound on a delivery is answered by moving freight between periods at the price of one period's holding cost, and so is a bound on a *single* period's dispatch total; the aggregate with no such escape is the horizon total. We therefore branch on the fleet, in two levels (Figure 5), keeping the delivery rung as a third entered only where the first two have nothing left to say — which, in the runs reported here, is nowhere: it is disabled throughout, as Section 9.1 records. Master, pricing problem, construction and tree are together the procedure we call BP-3D-SCPTOP (Algorithm 2).

Level 1: the total fleet. Given an incumbent with fleet vector $\mathbf{n}^\star$ and total $X^\star$, three regions partition the space. $X \geq X^\star + 1$ costs a whole truck with no offset and is usually pruned by bound; $X = X^\star$ fixes transport, so everything inside it is inventory allocation and this is where level 2 goes; and $X \leq X^\star - 1$ is the only region that can beat the incumbent outright, so it is worth a certificate of its own and the stack is ordered to expand it first. A plan found *inside* an inequality region splits it on its own total the same way, so level 1 recurses.

Level 2: the fleet vector, inside $X = X^\star$. Fix an order on the pairs (g, t). The fleet vector of any plan in the region differs from $\mathbf{n}^\star$ at a first index j, or nowhere, so branching there gives three children,

$$n_j \leq n_j^\star - 1, \qquad n_j = n_j^\star, \qquad n_j \geq n_j^\star + 1, \tag{81}$$

disjoint and exhaustive, with the equality child partitioned again at $j + 1$ (Murty's scheme (Murty 1968)) and expanded first. The partition is taken around the plan *constructed at that node*, so a node that improves the incumbent is re-partitioned around the better plan at once and nothing is restarted.

Pruning, and what counts as a proof. A node is a region imposed on the RMP as fleet rows. Its LP over the current pool is solved first as a filter (Section 6.4); where the filter allows pruning and the certification policy admits the node, column generation is run to a certificate, and a certified value at or above the incumbent prunes it while a value below becomes that region's bound. Every other node carries its parent's, valid because the child's region is a subset. An infeasible node relaxation is *not* a proof that a region is empty: stock non-negativity is not elastic, so a fleet lower bound can force deliveries up through whatever columns the pool holds and kill the relaxation for a reason internal to the pool. Only arithmetic licenses that sentence, and three such arguments are available: the fleet box intersected with the interval on X, an availability cap telescoped from the sending rows (11) and (13) under non-negative stock, and the constants of (59) read as a requirement floor.

What the tree reports. The incumbent is the cheapest verified plan any construction found. The global bound is the minimum over the *frontier*, not the root, a region nobody expanded contributing nothing; at the deadline the open regions are swept and certified where the filter allows. Under 3D loading a region's certificate comes, like the root's, from the volumetric relaxation with the region's fleet rows applied, which Proposition 2 covers because those rows bound the fleet vector and not the load model. A column, by contrast, is a feasible load, so one found in any region is valid in every other and a pruned node leaves its columns behind. Optimality is claimed *only* when the root is certified, no region remains open, no restart has broken the partition and the incumbent meets the global bound. The hierarchy is finite and its levels exhaustive, so those conditions are reachable in principle; no run here meets them.

Two levels of the three ran. The delivery level is implemented and was left disabled in every run reported here, so what covers a full-depth Murty leaf is a certificate rather than a further split — and a leaf whose certificate sits below the incumbent is then bounded but not refuted, however much budget remains. That is a property of the configuration and not of the procedure: with the delivery level on, such a leaf is split again. It is also not what stopped these runs, which ended with between six and fifteen regions still on the stack, i.e. on their budget and not on the hierarchy.

8.1 The initial pool

The search starts from the plainest possible pool: one maximally full truck per commodity and direction, obtained geometrically by pricing that commodity alone with every other capped at zero — so that where the probe closes, the pool truck is that commodity's placeable maximum and the same number is its placeability cap — and volumetrically by filling to capacity. No demand mixes and no reduced variants are seeded, so whatever the search reaches it reached by pricing, branching and construction.

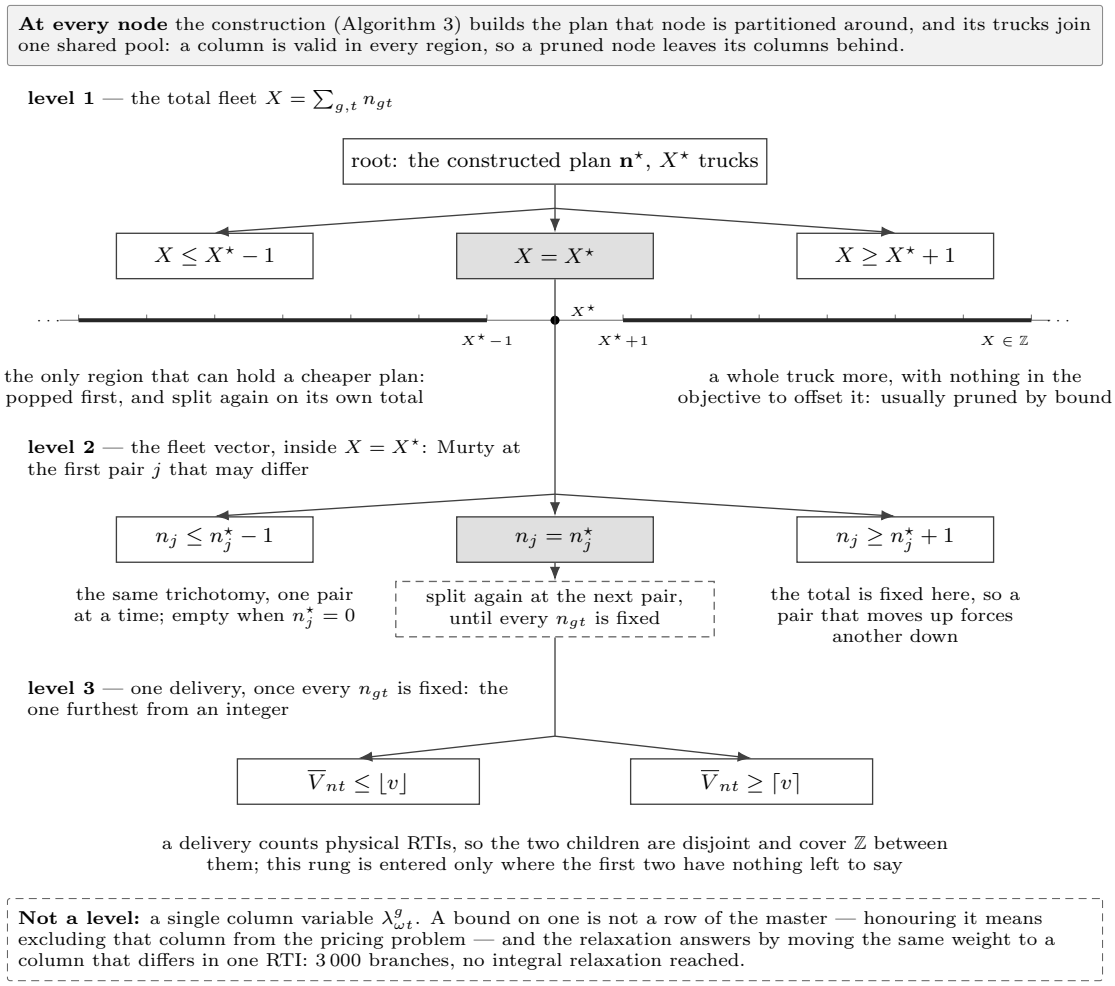


Figure 5: Every rung bounds an integer quantity the master already prices, so the children of a split are disjoint and together exhaustive and the search gives up nothing by taking them. The order is forced by the objective: transport costs $cK \cdot X$ and nothing else in the model counts trucks, so the horizon total is the one aggregate that cannot be answered by moving freight between periods for one period's holding cost, as a single dispatch count or a single delivery can. And because these are rows of the master and not restrictions on the subproblem, branching moves the duals and leaves the pricing problem alone — which is also how a load no dual vector would have asked for (Figure 4) comes to be priced at all.

8.2 The flow-first construction

The primal heuristic decides first how much moves and how many trucks carry it, and builds the trucks afterwards. It runs inside the region $\mathcal{R}$ of the node that calls it and must stay there, or the children that node induces would partition around a point it excludes, and it runs on a *fresh copy* of the restricted master, step 2 changing variable types the node's own LP needs intact. Figure 6 follows one period through the four steps of Algorithm 3.

1. *Restricted master.* The LP over the pool, inside $\mathcal{R}$, gives the duals and the fractional point.
2. *Aggregate integer master.* Introduce the dispatch totals n_{gt} as explicit integer variables, make $\overline{V}_{nt}, \overline{V^{\mathrm{e}}}_{rt}$ integer and leave the column variables *continuous*, with $\mathcal{R}$ imposed on the n_{gt} as well as on the columns and, under 3D loading, the stack-count rows (87) added. The objective is decided here, and *without a single call to a pricing problem*: the row set is the master's own. Artificial slack above a tolerance means the region is infeasible over this pool, and the node is abandoned.

Algorithm 2 BP-3D-SCPTOP: branch and bound over the fleet vector, with CONSTRUCT as primal heuristic and CERTIFY as bound.

$\bar{\Omega} \leftarrow$ one maximally full single-commodity truck per commodity and direction, each verified ▷ no demand mixes, no reduced variants
$\mathrm{LB}_0 \leftarrow \textsc{Certify}(\bar{\Omega}, \mathrm{root})$; $(\bar{z}, \mathbf{n}^\star, \bar{\Omega}) \leftarrow \textsc{Construct}(\bar{\Omega}, \mathrm{root})$; $X^\star \leftarrow \sum_{g,t} n^\star_{gt}$
$\mathcal{S} \leftarrow [\, X \geq X^\star + 1,\ X = X^\star,\ X \leq X^\star - 1\,]$ ▷ a stack: the region that can beat the incumbent is popped first
while $\mathcal{S} \neq \emptyset$ and time remains **do**
    $\mathcal{R} \leftarrow$ pop $\mathcal{S}$; impose it on the RMP as fleet rows
    if $\mathcal{R}$ is refuted arithmetically **then**
        continue ▷ fleet box, availability, requirement
    end if
    $\tilde{z} \leftarrow$ LP over $\bar{\Omega}$ inside $\mathcal{R}$ ▷ over-estimates the region's LP: a filter, never a bound
    if $\tilde{z} \geq \bar{z}$ and the policy certifies $\mathcal{R}$ **then**
        $(\mathrm{LB}_{\mathcal{R}}, \mathrm{proved}) \leftarrow \textsc{Certify}(\bar{\Omega}, \mathcal{R})$
        if $\mathrm{LB}_{\mathcal{R}} \geq \bar{z}$ **then**
            continue ▷ pruned
        end if
        record $\mathrm{LB}_{\mathcal{R}}$ as an open bound
    else
        $\mathcal{R}$ carries its parent's bound
    end if
    $(z, \hat{\mathbf{n}}, \bar{\Omega}) \leftarrow \textsc{Construct}(\bar{\Omega}, \mathcal{R})$ ▷ its columns join the shared pool whatever it returns
    if no plan **then**
        price under $\mathcal{R}$'s duals and construct once more; **continue** if still none
    end if
    if $z < \bar{z}$ **then**
        $(\bar{z}, \mathbf{n}^\star) \leftarrow (z, \hat{\mathbf{n}})$ ▷ new incumbent
    end if
    if $\mathcal{R}$ does not fix X **then**
        push the three regions of level 1 around $\sum_{g,t} \hat{n}_{gt}$ ▷ level 1 recurses
    else if the Murty depth j is below the cap **then**
        push the three children (81) at the j-th pair, around $\hat{\mathbf{n}}$, equality last
    end if
end while
for all $\mathcal{R}$ still on $\mathcal{S}$, while a reserve of time remains **do**
    filter by the pool LP; certify the survivors; record their bounds ▷ the frontier
end for
$\mathrm{LB} \leftarrow \max\{\mathrm{LB}_0, \min$ over the open and frontier bounds$\}$, capped at $\bar{z}$
verify every dispatched truck of the incumbent against (48)–(53)
return $(\bar{z}, \mathrm{LB})$; optimality is claimed only if LB_0 was proved, $\mathcal{S} = \emptyset$, no restart occurred and $\bar{z} = \mathrm{LB}$

3. *Partition, as a cover.* For each direction and period, realise the integer deliveries $Q^\star_{gt}$ as $n^\star_{gt}$ legal

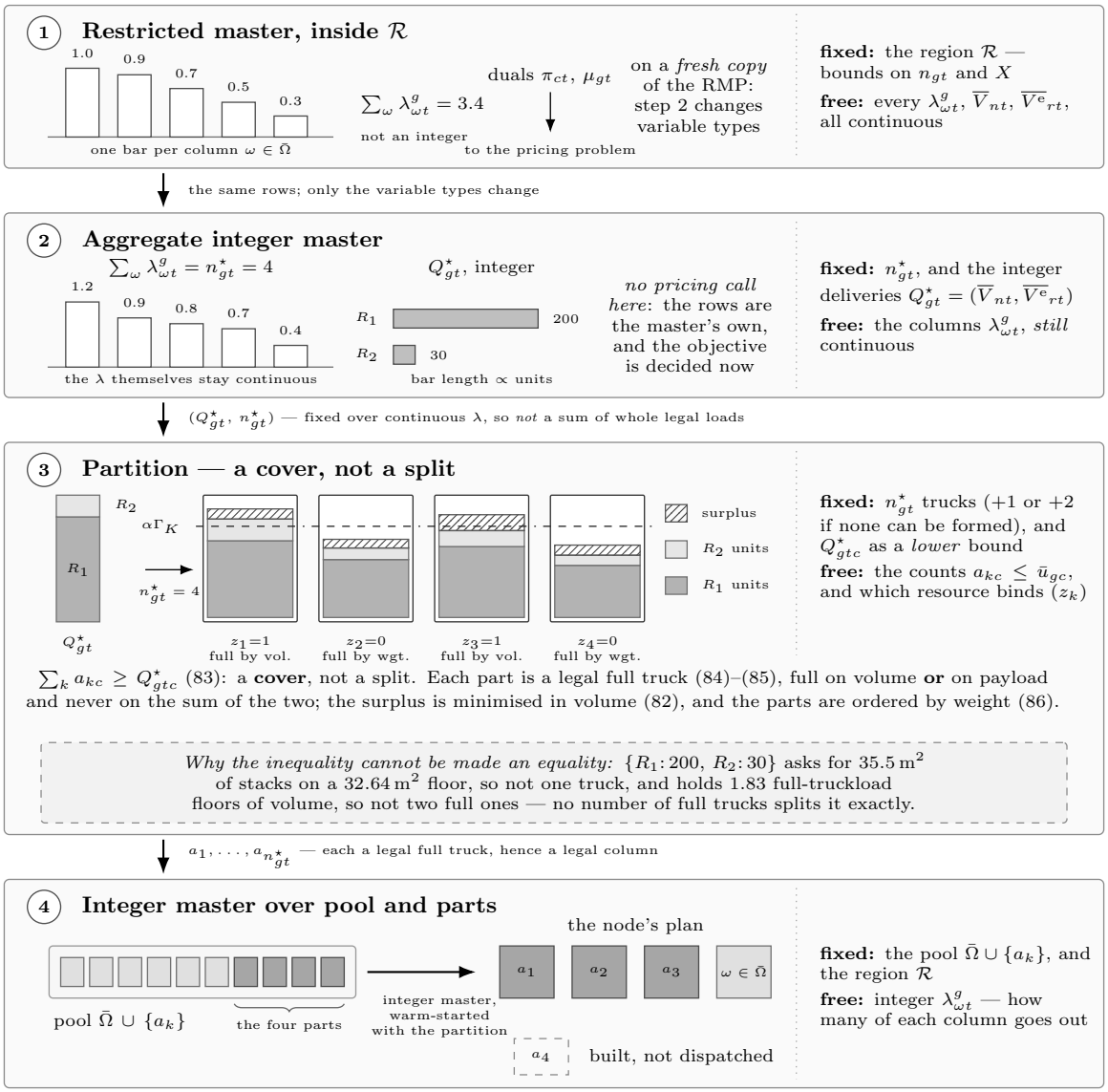


Figure 6: The flow-first construction over one period, and why its third step covers rather than splits. Steps 1 and 2 settle how much moves and how many trucks carry it while the column variables stay *continuous*, so the delivery vector they fix need not be a sum of whole legal loads — for the vector shown, no number of full trucks realises it exactly. Step 3 therefore asks only that the parts *cover* it (83), paying for the hatched surplus (82), and makes each part full on volume *or* on payload (85) rather than on the sum of the two. Nothing built is binding: the parts enter step 4 as columns with a warm start, and the integer master may dispatch fewer of them than the cover produced.

trucks by an integer programme over per-truck commodity counts a_{kc}, $k \le n^\star_{gt}$:

$$\min \quad \sum_c \gamma_{gc}\Big(\sum_k a_{kc} - Q^\star_{gtc}\Big) \tag{82}$$

$$\text{s.t.} \quad \sum_k a_{kc} \ \ge \ Q^\star_{gtc} \qquad \forall c \tag{83}$$

$$\textstyle\sum_c \gamma_{gc} a_{kc} \le \Gamma_K, \qquad \sum_c w_c a_{kc} \le pK^w \qquad \forall k \tag{84}$$

$$\textstyle\sum_c \gamma_{gc} a_{kc} \ge \alpha\Gamma_K \theta_k, \qquad \sum_c w_c a_{kc} \ge \alpha p K^w (1-\theta_k) \qquad \forall k \tag{85}$$

$$\textstyle\sum_c w_c a_{kc} \ \ge \ \sum_c w_c a_{k+1,c} \qquad \forall k < n^\star_{gt} \tag{86}$$

$$a_{kc} \in \{0, \ldots, \bar{u}_{gc}\}, \qquad \theta_k \in \{0,1\}.$$

(84)–(85) make each part a legal truck, θ_k selecting the resource on which part k meets the floor; (86) orders the parts by weight; $\bar{u}_{gc}$ is the placeability cap below. If $n^\star_{gt}$ parts cannot be formed, $n^\star_{gt}+1$ and $+2$ are tried where $\mathcal{R}$ allows — and a timeout is not a proof that none can be.

4. *Integer master over pool plus parts.* The parts join the pool as columns and the integer master is solved over it, warm-started with the partition; its solution is the node's plan.

Why the partition must be a cover. This is the step at which the procedure could quietly become wrong. With integer λ the aggregate $\sum_\omega b_{\omega n}\lambda^p_{\omega t}$ is a sum of placeable loads, so an unrealisable delivery vector cannot be requested; step 2 breaks that link by fixing the aggregates over *continuous* λ. A fractional blend of legal loads need not be realisable as legal loads for *any* number of trucks: floor area is consumed in whole footprints while (85) is measured in volume, so splitting a vector into more trucks consumes more floor without adding volume and splitting it into fewer violates the trailer. On one benchmark instance $\{R_1\!:\!200, R_2\!:\!30\}$ holds 1.83 full-truckload floors of volume but needs $35.5\,\mathrm{m}^2$ of stacks on a $32.64\,\mathrm{m}^2$ floor. A cover is sound because every part is a legal column and the parts enter step 4 *as columns with a warm start*: the integer master may dispatch fewer than were built, and over-delivering is feasible, stock having no upper bound.

Algorithm 3 CONSTRUCT$(\bar{\Omega}, \mathcal{R})$ — the flow-first relax and fix. Lines 9–14 run under 3D loading only. ε_2 is step 2's relative MIP gap and *tol* the tolerance above which artificial slack means the region admits no plan over this pool.

1: $\bar{\Omega}' \leftarrow$ a fresh copy of the RMP over $\bar{\Omega}$ ▷ not the node's own LP: step 2 changes variable types
2: solve LP$(\bar{\Omega}', \mathcal{R})$ ▷ duals π, μ and the fractional point the node stands on
3: make $n_{gt}, \overline{V}_{nt}, \overline{V^{\mathrm{e}}}_{rt}$ integer; keep λ continuous; impose $\mathcal{R}$ on n_{gt} and on λ
4: add the aggregate form of (87) ▷ 3D only; layer coefficient = the height count, the crush limit being its own row
5: solve to relative gap ε_2
6: **if** no solution, or artificials $>$ tol **then**
7: **return** $\perp$ ▷ node abandoned; the region is infeasible over this pool
8: **end if**
9: $(Q^\star, n^\star) \leftarrow$ the integer deliveries and fleet
10: **for all** $(g,t) \in G \times T$ **do**
11: **for** $m = n^\star_{gt}$, $n^\star_{gt}+1$, $n^\star_{gt}+2$ while $\mathcal{R}$ allows **do**
12: solve PARTITION$(Q^\star_{gt}, m)$, (82)–(86), with $\bar{u}_{gc}$ as variable bounds
13: **break** if feasible ▷ a timeout is not a proof of infeasibility
14: **end for**
15: **for all** parts a_k of the cover **do**
16: (verdict, layout) $\leftarrow$ PACK(a_k)
17: **if** placeable **then**
18: verify the layout; $\bar{\Omega} \leftarrow \bar{\Omega} \cup \{(a_k, \text{layout})\}$
19: **end if**
20: **if** *proved* unplaceable **then**
21: bisect $\sigma \in (0,1]$ and cut every load dominating $\lfloor \sigma a_k \rfloor$ ▷ validity
22: **end if**
23: **if** undecided **then**
24: exclude a_k as a single point ▷ heuristic restriction, never validity
25: **end if**
26: **end for**
27: re-solve the partition if a cut was written and rounds remain
28: **end for**
29: $\bar{\Omega} \leftarrow \bar{\Omega} \cup \{\text{the parts}\}$; solve the integer master over $\bar{\Omega}$, warm-started with the partition
30: **return** $(z, \hat{\mathbf{n}}, \bar{\Omega})$ ▷ the parts are kept even when no plan is returned

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8.3 The construction under 3D loading

Rows (84)–(85) are exactly the volumetric column conditions, so over that oracle every part is a column by construction. Under 3D loading a part must also satisfy (48)–(53); since the volumetric conditions are *necessary* for geometric feasibility (Proposition 2) the partition programme is kept as a *relaxation* of the geometric one, and each part is then tested for placeability. Step 2 makes no oracle call, so the objective is still decided before any geometry is examined.

Everything here rests on one property. Placeability is **downward closed**: given a layout of $a \geq b$, removing the surplus units from their slots leaves positions unchanged, so nothing overlaps, every stack is no taller and no heavier, and an emptied slot is dropped. The full-truckload floor is no part of placeability, so it does not disturb the closure.

Deciding placeability. A part is submitted to an oracle that returns one of three answers. A constructive *shelf design* — a small integer programme laying the part's stacks in shelves along the trailer, one integer count per shelf, footprint and orientation — supplies essentially every positive answer, and its layout passes the independent verifier of Section 9.1 before it counts. The pricing problem, in feasibility mode with the true slot bound (46), supplies the *infeasibility* proofs.

Two kinds of cut, which must never merge. A part *proved* unplaceable yields a *dominance cut* forbidding every load that dominates it componentwise, valid by the closure above, so one cut removes the whole up-set of the refuted load — including every point of the one-unit-at-a-time staircase the partition would otherwise walk down. Before cutting, the load is tightened along its own ray: $\lfloor \sigma a \rfloor$ is probed for σ bisected in $(0, 1]$ and the cut written at the smallest scaled point still *proved* unplaceable. A part left *undecided* is instead excluded as a single point so that the partition proposes something else: a restriction of a primal heuristic, never a validity claim. Reading an undecided verdict as a dominance cut is how a method comes to forbid its own optimum.

Placeability caps. The largest number of one commodity a truck can hold is a scalar and, by the same closure, monotone, so it is computed rather than walked to: bisected once per direction and commodity at the start of a run, a proved optimum tightening it and an undecided probe leaving the analytic cap, then handed to the partition as $\bar{u}_{gc}$.

A stack relaxation inside the partition. The partition additionally carries, per truck k and footprint class f, an integer number of stacks $s_{kf} \leq \lfloor pK \cdot qK / p_f q_f \rfloor$, the crush limit and the height budget imposed per stack and the stacks' areas summed against the floor:

$$\sum_{c \in f} w_c a_{kc} \leq W^{\max} s_{kf}, \qquad \sum_{c \in f} r_{gc}\, a_{kc} \leq rK\, s_{kf}, \qquad \sum_{f} p_f q_f\, s_{kf} \leq pK \cdot qK, \tag{87}$$

and step 2 carries the same rows per (direction, period) with the area right-hand side multiplied by n_{gt}. Both are necessary, every truck of the period having integer stack counts that satisfy the per-truck rows, and they are what the volumetric relaxation omits. Where a class carries one RTI type the height row sharpens to $\sum_{c \in f} a_{kc} \leq \lfloor rK / r_{gr} \rfloor s_{kf}$: the crush limit is already carried by the first row with each commodity's own weight, so the height row takes the height-limited count, which is what the runs reported here used; the smaller of the two counts is also valid, and strictly tighter. These aggregates remain a relaxation of the per-truck split, which is why step 3 keeps its $n + 1$ fallback.

9 Computational results

9.1 Protocol, instances and what each quantity means

The masters, the partitions, the volumetric benchmark and the compact models are solved with Gurobi 12; the pricing problems and the placeability tests with CP-SAT from OR-Tools 9.12. Table 4 declares, per stage, the five parameters that can change a solver's answer: time limit, MIP gap, threads, per-model memory cap and seed. Memory is part of the budget — Gurobi holds its tree in memory — so the time at which a fixed cap is reached licenses no inference about difficulty. Every row below is read from a recorded run, each one declaring its own budget, thread count, memory cap and commit. A *gap* is always $(z - \mathrm{LB})/z$ against the method's own certified bound; Table 5 defines the rest.

Every row is one run, and one run is reproducible: the procedure is deterministic given a build, a machine and the way its code is executed. That was checked rather than assumed, and the check is worth reporting in full. On $I_16_15_2_90_k25$ in 3D, three runs with the solver's module compiled in the running process returned the same plan at the same cost to the cent, and two runs of the same source from its precompiled form also agreed to the cent — with each *other*, at a different plan: 55 trucks at 79 608.45 against 59 at 83 645.86. Nothing about the model or the configuration differs between them; only how fast the machine executes the same instructions.

That is what a wall-clock budget costs. Every phase of the procedure draws on one, so which phases finish, how many construction attempts fit, and which regions the tree opens all depend on execution speed, and a plan is the method's answer under one schedule rather than a function of its configuration. The effect is largest where a phase ends near its deadline: on the largest ladder instance the two execution paths differ by whether a plan is found at all. Every run reported here was executed the same way, so the table is internally comparable, and two consequences are taken seriously throughout: a difference of one or two trucks between rows is not read as an effect, and no claim rests on one.

Three settings that are not budgets also decide what the tables can say. Column generation is run to a certificate at the root and at the one region that can refute the incumbent outright — "is there a plan with one truck fewer" — and nowhere else, which is the policy every run reported here used. Certifying every node instead is affordable on the volumetric model and closes far more of the space there, so the *open* counts below are a property of that choice and not of the method; it is not affordable under 3D loading, where one geometric sweep costs minutes, and the two modes are reported under one policy so that their columns mean the same thing. The second is that the branching hierarchy's third level — the deliveries — is disabled in every run reported here, so a full-depth leaf is covered by a certificate rather than split again; Section 8 says what that costs and what it does not. The third is that no seed is set by the implementation of Table 6 and the size ladder; nothing reported rests on one.

Table 4: Declared budgets, one line per stage that produces a number reported below. *gap* is the relative MIP gap at which an integer solve may stop and *mem* the per-model soft cap. The two placeability censuses differ in oracle as well as in limit: the thirty-six instances were put to the constraint programme alone, the size ladder to a constructive shelf witness with the constraint programme only for refutations, which is why only the first leaves loads undecided. Every line is read from the run files themselves. The volumetric parity runs declare their own configuration in each row, which is what lets Table 6 compare like with like; the runs they replace recorded none, and a thread count asked for on the default Gurobi environment had not reached the models, so those used all twelve cores of the machine. No seed is set by that implementation, so the solver's default applies — and no number reported here rests on a seed, every stage being either solved to proved optimality or stopped on a declared budget.

Stage	solver	time (s)	gap	threads	mem (GB)	seed
BP-3D-SCPTOP, 36 instances	Gurobi + CP-SAT	900	—	10	20	0
BP-3D-SCPTOP, size ladder	Gurobi + CP-SAT	900–3 600	—	10	20	0
Compact SCPTOP (volumetric)	Gurobi	600–3 600	0	10	20	0
Compact 3D-SCPTOP, $n_{\mathrm{LC}} = 3$	Gurobi	1 800	0	10	20	0
Fleet-floor certification	Gurobi	to convergence	—	1	20	0
Placeability census, 36 instances	CP-SAT	300 per load	—	—	—	—
Placeability census, size ladder	shelf + CP-SAT	60 per load	—	—	—	—
BP-3D-SCPTOP, volumetric parity	Gurobi	3 600	—	10	20	not set

Instances. Thirty-six instances form a full factorial over the products $|N| \in \{4, 8\}$, the horizon $|T| \in \{5, 10\}$, the RTI types $|R| \in \{2, 3, 4\}$ and the volumetric granularity $v \in \{60, 90, 120\}$, named $I_|N|_|T|_|R|_v$, where v is the number of erected RTIs that fill one truck by volume, so larger v means smaller RTIs and a harder placement subproblem. Footprints are the five of the industrial catalogue, from 800×600 to 2400×1200 mm, with heights quantised to 50 mm and bounded so that at least two layers fit under the roof; from $|R| \geq 3$ several types share a footprint, which lets a stack mix types across layers. Demand deviates $\pm 10\%$ from a mean of roughly 2.5 truckloads per period, and content weight by $\pm 25\%$ between products sharing a type, which is what makes the crush limit depend on the period's mix. The truck is a semi-trailer, $13600 \times 2400 \times 3000$ mm, payload 24 t, $W^{\max} = 1200$ kg, $\alpha = 0.55$, $cK = 1000$ per dispatch, $lt = 1$, one period of coverage stock. A size ladder follows the same rules over $(|N|, |T|) \in \{(8, 10), (12, 12), (16, 15), (24, 20)\}$

Table 5: What each reported quantity means. A sentinel recorded by a run that obtained no bound is never printed, never averaged and never rounded to a side.

z	The cost (54) of the plan returned, net of any artificial penalty; a plan is reported only if it carries no artificial slack. An upper bound on the optimum in every case.
trucks	$\sum_{g,t} n_{gt}$ for that plan.
LB	A lower bound on the optimum of the model the method solves. For the method it is the *certified* bound of Section 6.4, printed only where the sweep behind it came back with every pair proved; where it did not, the row says *unproved* and carries no gap. For a compact model it is the solver's dual bound at its stop.
gap	$(z - \mathrm{LB})/z$, against the method's own bound. An optimality gap only to the extent that bound is tight.
stop	How a solve ended: OPTIMAL, TIME_LIMIT or MEM_LIMIT. A time or memory stop makes the row's time the budget rather than a solve time.
nodes / open	Tree nodes opened; regions left unexpanded at the deadline. A run with open regions has not searched the space, whatever its gap — and a run with none has not refuted it either, a region closed with a bound below the incumbent being bounded and not excluded.
verified	Dispatched trucks re-derived from the column's own fill and found legal by the independent routine of Section 9.1, out of the number dispatched.
unpl. / und.	Among the loads a *volumetric* plan dispatches, those *proved* unplaceable and those left *undecided* within the limit; an undecided load is counted on neither side.
Δz	The cost of one method relative to another on the same instance, with the denominator named where it appears.

crossed with the truckloads consumed per period, 2.5, 5 or 8, at $|R| = 2$ and $v = 90$; its name's suffix is that figure times ten, not K.

Verification, and a second implementation. Every plan below is re-derived from its dispatch list alone and checked truck by truck against (48)–(53) by a routine independent of the search and of the oracle that built the column, re-deriving each slot's weight from the fill the slot carries and rejecting a fill that does not describe the stack it sits in. It shares the full-truckload predicate with the oracle and the partition floor, so it catches a defect in a solver's *satisfaction* of that definition and not one in the definition. The method has in addition been re-implemented in a second language and solver stack and checked layer by layer against the first, agreeing on $S_2_2_2$ and $I_4_5_2_60$ on every priced column, on every master dual to 0.00e+00, on the certified bound to the digit and on the whole search trajectory of the instance where the search exhausts — agreement that is partly by construction, a port having been tested against the reference, and on $I_4_5_2_60$ the two plans in fact differ.

Which of the two produced which table is stated here rather than left to be inferred. Table 6 is the second implementation, on the equal budgets its caption declares. Every other table is the first. The two are not interchangeable at the sizes of the size ladder: column generation there runs as the ladder of Section 7, a heuristic rung ahead of the exact oracle, and the second implementation has only the exact rung. Where the heuristic rung contributes nothing — the three smallest ladder rungs, measured below — the two agree or the second is slightly ahead; where it contributes most of the trucks, the second implementation is behind on one rung and returns no plan on the largest. A certificate is unaffected either way, an exact sweep that returns nothing being a certificate on its own, so this is a statement about which plans each can find and not about what either may claim.

9.2 What the volumetric side gives

The volumetric problem of Appendix A earns its place in this study for one reason: it is the only part of the problem for which an independent optimum exists, because a compact formulation of it can be solved exactly on the smaller instances. What Table 6 measures is therefore not which method wins but whether the procedure arrives where it should on a case whose answer is known. Both sides were given the same 3 600 s, 20 GB and ten threads.

On the two instances the compact model closes, it does. On $I_8_5_2_90$ the procedure returns 18 796.96 with twelve trucks — the proved optimum, to the cent, reached from a different formulation by a different

Table 6: BP-3D-SCPTOP under its *volumetric* oracle against the compact SCPTOP formulation, both solving the *volumetric* problem of Appendix A, so that the two bracket one optimum. **This table is a check on the procedure, not a contest**: the volumetric problem is the one case here where an independent optimum exists, and the question each row answers is whether the procedure reaches it. z is the incumbent at the stop; OPT marks a proved optimum, ^{m} a stop on the memory cap, ^{t} a stop on time. "bound" is Gurobi's dual bound for the compact model and the certified master-LP bound of Section 6.4 for the method. Δz is $(z_{\mathrm{DW}} - z_{\mathrm{compact}})/z_{\mathrm{compact}}$; negative means the method's plan is cheaper. Both sides had 3 600 s, 20 GB and ten threads — except that the compact model was given 1 800 s on *I*_8_5_2_90, where it proved the optimum in 0.2 s and the budget never bound. The method's wall clock stops short of 3 600 s by construction: the search is given $(1-0.15)\times 3\,600 = 3\,060$ s and the rest is reserved for the frontier sweep that bounds the regions left open, which returns here in 2 to 73 s rather than spending its reserve. Reserved out of the budget and not on top of it — the recorded runs this table replaces finished at 1 817 to 1 897 s against an 1 800 s budget, and a row whose runtime exceeds the budget it is compared at is not a comparison. No plan in this table is verified truck by truck: a volumetric column carries no layout.

	compact SCPTOP				BP-3D-SCPTOP (volumetric oracle)					
Instance	z	bound	trucks	t (s)	z	bound (cert.)	trucks	t (s)	Δz (%)	tighter bound
*I*_8_5_2_90	18 796.96 OPT	18 796.96	12	0.198	18 796.96^{t}	18 795.56	12	3064	0.0000	18 796.96 (compact)
*I*_8_10_2_90_*k*25	38 855.65^{m}	38 851.93	26	3269	38 853.17^{t}	38 791.19	26	3062	−0.0064	38 851.93 (compact)
*I*_16_15_2_90_*k*25	68 139.17 OPT	68 139.17	45	1350	68 139.21^{t}	68 131.48	45	3117	+0.0001	68 139.17 (compact)
*I*_16_15_2_90_*k*50	134 346.91^{m}	133 417.47	89	1122	134 507.47^{t}	133 484.56	89	3073	+0.1195	133 484.56 (DW)
*I*_24_20_2_90_*k*50	200 266.21^{m}	199 518.04	137	2093	200 048.69^{t}	200 013.57	137	3071	−0.1086	200 013.57 (DW)
*I*_24_20_2_90_*k*80	282 101.77^{m}	280 823.53	189	2403	281 233.28^{t}	280 910.22	188	3133	−0.3079	280 910.22 (DW)

search. On *I*_16_15_2_90_*k*25 it returns 68 139.21 against a proved 68 139.17, also at forty-five trucks: four cents on sixty-eight thousand, 0.0001% above the truth, with its own certified bound 0.0113% below it. The procedure brackets a known optimum from both sides to within about a hundredth of a percent. That is the check this subsection exists to report, and it passes.

Where the compact model works it is also far the faster, at 0.2 s and 1 350 s against the procedure's roughly 3 070 s. On a volumetric problem small enough to write down whole, writing it down whole is the better plan; the decomposition is built for the problem the compact model cannot express, and Section 9.3 is where that is asked of it.

On the four instances where the compact model exhausts its memory cap with its dual bound short of its incumbent, the two agree closely enough to say the procedure is solving the same problem well: costs within 0.31% on every row, the same fleet on three and one truck fewer on *I*_24_20_2_90_*k*80. The procedure's plan is cheaper on three and dearer on one, by 0.1195% on *I*_16_15_2_90_*k*50, the row where it opened twelve nodes in the hour and left its own interval 0.76% wide. At this accuracy the two are measuring each other, and neither row is a ranking.

Where the decomposition does separate itself is the bound: on the three largest instances its certificate is the tighter of the two, by 67, 496 and 87 cost units. The compact model's dual bound degrades as its memory cap approaches, while the master LP over a finite pool certifies whatever the pool supports.

One such statement needs no comparison of plans at all. On *I*_12_12_2_90_*k*25 the method certifies 52 105.31 — a full sweep in which every pair priced optimal and nothing priced out, over the volumetric relaxation and so valid for the 3D problem by Proposition 2 — 274 s into its run, where the compact volumetric model still stood at a dual bound of 51 944.22 when its 600 s expired, both at ten threads and a 20 GB cap. Both bound the same relaxation, and the decomposition's is 0.310% tighter.

9.3 What the 3D side gives

On the thirty-six generated instances at a uniform 900 s the method returns a verified plan on thirty-one and a proved bound on thirty-five, at a mean certified gap of 8.8% over the thirty rows with both. The five failures are not spread across the factorial: all five have $v = 60$, where one container is the largest fraction of a trailer and a coarse integer loading structure leaves the construction least room to cover a delivery vector with full trucks — which is also why a volumetric capacity is least adequate there (Section 9.5). On those five rows *no plan* is returned rather than an invalid one, which follows from the formulation: a column the verifier would reject never enters the pool.

The ladder takes the same method to twenty periods and 160 trucks under the same guarantee. On

Table 7: BP-3D-SCPTOP on the size ladder, every row on one commit (73411b9), ten threads, a 20 GB per-model cap and seed 0, run sequentially. *root* is the plan the construction returns before any node is popped and *tree* the improvement the search then makes on it, both read from the same run's trace, so what branching contributes is measured within a run rather than across commits. *open* is the regions still on the stack at the deadline and *dropped* those left with no bound covering them; both 0 does *not* mean the space was refuted, a region closed with a certified bound below the incumbent having been bounded and not excluded, which is why row 1 claims no optimum of its own. The bound is the certified volumetric relaxation, valid for 3D by Proposition 2; it is loose wherever the crush limit binds, so the gap on the last two rows is not an optimality gap. LB and *gap* are the *method's own* certificate on every row, never a mixture: on rows 2 and 4 the compact volumetric model supplies a tighter valid bound (38 851.93 at a memory stop, 68 139.17 proved optimal), which Table 6 reports and Table 10 uses as an interval end. Row 1 is the paper's only proved optimum and the proof is not the method's: the compact volumetric relaxation of that instance solves to 18 796.96 in 0.198 s and this verified 3D plan attains it. Row 1's instance is also row $I_8_5_2_90$ of Table 12: **two runs, on two commits** (there 0c0c16c at 900 s), agreeing on z, trucks and bound and differing at 68 nodes against 67 here, with the gap printed there to two decimals rather than four. The twelve columns are set as two blocks of the same five rows in the same order, so that the table needs no reduction in type size.

Instance	z	trucks	verified	LB (cert.)	gap
$I_8_5_2_90$	**18 796.96**	12	12 / 12	18 795.54	0.0076%
$I_8_10_2_90_k25$	38 855.34	26	26 / 26	38 791.14	0.165%
$I_12_12_2_90_k25$	52 129.45	34	34 / 34	52 105.25	0.046%
$I_16_15_2_90_k25$	78 549.14	54	54 / 54	68 131.43	13.3%
$I_24_20_2_90_k50$	230 594.36	160	160 / 160	200 013.41	13.3%

Instance	root (t)	tree	nodes	dropped	open	wall / budget (s)
$I_8_5_2_90$	18 796.96 (140 s)	0.00%	67	0	0	873 / 900
$I_8_10_2_90_k25$	38 865.30 (136 s)	0.03%	68	0	35	883 / 900
$I_12_12_2_90_k25$	53 195.84 (459 s)	**2.00**%	11	0	10	1 761 / 1 800
$I_16_15_2_90_k25$	78 549.14 (1 181 s)	0.00%	1	0	2	1 746 / 1 800
$I_24_20_2_90_k50$	230 594.36 (1 418 s)	0.00%	2	0	4	3 492 / 3 600

Table 8: Fleet floors certified offline. The volumetric relaxation's master, restricted to $\sum_{g,t} n_{gt} \le X$, is priced to convergence on one thread; INFEASIBLE means no plan with at most X trucks exists, in 3D as well, since every placeable load is volumetrically feasible and the fleet row does not depend on the load model (Proposition 2). *time* is the region certification alone; the root certificate it rests on took 4 to 32 s. These are the only proofs of minimality in this paper.

Instance	plan's trucks	X tested	verdict	time (s)
$I_8_5_2_90$	12	≤ 11	INFEASIBLE: 12 is the minimum fleet	1.5
$I_8_10_2_90_k25$	26	≤ 25	INFEASIBLE: 26 is the minimum fleet	3.9
$I_12_12_2_90_k25$	34	≤ 33	INFEASIBLE: 34 is the minimum fleet	16.3

Table 9: Table 12 by instance factor. *plans* counts the instances on which BP-3D-SCPTOP returned a verified plan inside the 900 s budget; the mean gap is over those instances and over the rows whose bound was proved, and the count of each is given so that the denominator is never in doubt. The last three columns are the volumetric benchmark's dispatched loads: the share a constraint programme *proves* unplaceable, the share it leaves *undecided*, and the share it places. The proved share is a *floor*, since more oracle effort can only move loads out of the undecided column.

factor	value	plans	mean gap (rows)	loads	unpl.	und.	placed
$\|N\|$	4	16 / 18	8.9% (16)	363	67.2%	26.4%	6.3%
	8	15 / 18	8.6% (14)	361	71.2%	21.6%	7.2%
$\|T\|$	5	17 / 18	8.4% (17)	219	56.2%	33.8%	10.0%
	10	14 / 18	9.4% (13)	505	74.9%	19.8%	5.3%
$\|R\|$	2	10 / 12	9.7% (10)	246	79.3%	15.4%	5.3%
	3	11 / 12	8.4% (10)	235	61.7%	29.4%	8.9%
	4	10 / 12	8.3% (10)	243	66.3%	27.6%	6.2%
v	60	7 / 12	19.3% (6)	182	91.8%	2.7%	5.5%
	90	12 / 12	5.0% (12)	239	55.2%	36.4%	8.4%
	120	12 / 12	7.4% (12)	303	66.7%	27.1%	6.3%
all		31 / 36	8.8% (30)	724	**69.2**%	24.0%	6.8%

the first three rungs the truck count is in addition the minimum possible, the region with one truck fewer being certified infeasible in 1.5, 3.9 and 16.3 s (Table 8), so what remains of those gaps, 0.0076%–0.165%, is inventory cost. The jump to 13.3% on the last two rungs is attributed to the bound and not to the plan, on the mechanistic evidence of Section 9.5; nothing here proves that attribution, and those two plans are reported as verified upper bounds under a stated budget. The claim about the gaps is therefore conditional: they are small where the volumetric relaxation is nearly exact, which is checkable in advance.

What branching contributes is uneven and is not averaged away. Read within one run (Table 7), the tree improves the root construction by 2.00% on the third rung, five node improvements taking 53 195.84 to 52 129.45, and by 0.03% or nothing on the other four: the pattern follows what the root leaves the tree, and where the relaxation is nearly exact the construction is already there. That the two generators are complementary is measured under one dual trajectory — the exact oracle's columns carry 4.72 distinct commodities on average against the heuristic rung's 10.17, which is Figure 4 in numbers — and in production that rung contributes 0 trucks to each of the three plans whose gap is under 0.2%, 0 of 12, of 26 and of 34, against 44 of 54 and 122 of 160 on the two where the crush limit binds.

The bound is cheap, which makes the division of labour of Section 6.4 affordable: over the forty-one runs reported here the volumetric sweep returns in 2.5 to 111.8 s with every pair proved on forty, the exception being the one row of Table 12 that prints *unproved* and is excluded from every mean. The geometric sweep certified no region on any run. Where the time goes depends on whether the crush limit binds: column generation takes 864 s of the 873 s first rung with the geometry free, while on the two crush-binding rungs the partition and its placeability calls dominate. Over the thirty-six instances the shelf design placed 24 686 loads and the constraint programme 246, which proved 4 327 of the 11 866 distinct compositions proposed unplaceable and left 397 undecided.

9.4 The compact model

A compact formulation of 3D-SCPTOP cannot serve as a yardstick beyond five periods. Written with a catalogue of $n_{\mathrm{LC}} = 3$ layouts shared by the trucks of a direction — a *restriction* of Section 4 chosen a priori, so its dual bound bounds the restriction and not $z^\star$ — it returns a plan on one of four ladder instances. On $I_8_5_2_90$, at 67 674 variables and 92 092 constraints, it stops on its 1 800 s limit with no phantom freight and twelve loads that pass the independent verifier, 0.102% above the method's plan at twice the method's budget. From ten periods on, at 119 424 to 322 734 variables, it returns *no feasible solution at all* in 1 800 s, all three stopping on time and not on memory — peak resident sets of 6.1, 2.7 and 4.5 GB against a declared 20 GB cap. Four rungs are four rungs; beyond five periods the compact model supplies neither a plan nor a bound that bounds the problem.

$I_8_5_2_90$ is also the one instance on which this paper reports an optimum, and the proof is not the

Table 10: The compact 3D-SCPTOP model (a catalogue of $n_{\mathrm{LC}} = 3$ layouts shared by the trucks of a direction, 1 800 s) against BP-3D-SCPTOP on the size ladder, the method's rows being those of Table 7. **The budgets differ on three rows**: the compact model had 1 800 s throughout, the method 900 s on the first two rows, 1 800 s on the next two — which are therefore the equal-budget pairs — and 3 600 s on the last. *none* means no feasible solution was found inside the budget, and all three such runs stopped on *time* with peak resident sets of 6.1, 2.7 and 4.5 GB against the declared 20 GB cap. The last column is the interval in which $z^\star$ lies, taking the method's plan as the upper end and, as the lower end, the tighter bound *that bounds the problem*: $n_{\mathrm{LC}} = 3$ is an a-priori restriction of Section 4, so the compact 3D model's dual bound bounds that restriction and not $z^\star$ and appears in its own column only, while a compact *volumetric* dual bound is valid for 3D by inclusion and is used instead where one exists. Row 1 is the one optimum (Table 7).

	compact 3D-SCPTOP, 1 800 s				BP-3D-SCPTOP					
Instance	z	trucks	LB	stop	z	trucks	LB (cert.)	gap	budget (s)	$z^\star$ in
$I_8_5_2_90$	18 816.13	12	18 795.76	TIME	18 796.96	12	18 795.54	0.0076%	900	[18 796.96, 18 796.96]
$I_8_10_2_90_k25$	*none*	—	36 458.75	TIME	38 855.34	26	38 791.14	0.165%	900	[38 851.93, 38 855.34]
$I_12_12_2_90_k25$	*none*	—	48 516.86	TIME	52 129.45	34	52 105.25	0.046%	1 800	[52 105.25, 52 129.45]
$I_16_15_2_90_k25$	*none*	—	64 053.23	TIME	78 549.14	54	68 131.43	13.3%	1 800	[68 139.17, 78 549.14]
$I_24_20_2_90_k50$	not run				230 594.36	160	200 013.41	13.3%	3 600	[200 013.41, 230 594.36]

Table 11: What volumetric planning costs, constructed rather than estimated: trucks dispatched on the two instances where the per-stack crush limit binds. Row 1 is the volumetric benchmark's own plan, which is not executable — 41 of its 45 dispatched loads on the first instance and 119 of its 137 on the second are *proved* unplaceable with *none* left undecided, so those shares carry no undecided caveat. Row 2 splits each unplaceable load where it stands, with no consolidation; it is reported for contrast and is *not* the comparison, consolidation being worth twelve and forty-four trucks. Row 3 is the comparison: the volumetric plan's delivery quantities kept, each period re-packed with the same partition-and-pack machinery the method uses, every resulting truck verified, and at least the original quantities delivered — by 80 and 137 units, the partition being a cover. Row 4 is BP-3D-SCPTOP, which also chooses the quantities. The volumetric plan on $I_24_20_2_90_k50$ is a *memory stop* and not an optimum, so a better one might repair into fewer than 212 trucks and that column's margin is overstated by an unquantified amount in the method's favour; the $I_16_15_2_90_k25$ plan is the proved volumetric optimum and carries no such caveat.

	$I_16_15_2_90_k25$	$I_24_20_2_90_k50$
volumetric plan (not executable)	45	137
repaired load by load (contrast only)	86	256
re-packed, quantities kept	**74**	**212**
BP-3D-SCPTOP (which also chooses the quantities)	**54**	**160**
saving on trucks / on cost	27.0% / 19.1%	24.5% / 16.2%

method's. The compact *volumetric* relaxation solves it to proved optimality at 18 796.96 in 0.198 s, every placeable load is volumetrically feasible, and the verified 3D plan attains that value; the method's own attempt failed: geometric certification of the three regions its search could not refute returned 18 795.57, 18 795.54 and 18 795.54, none of the three *proved*. The proof is available because this instance's two container families have coinciding height- and crush-limited layer counts — exactly the condition under which Section 9.5 finds a volumetric capacity adequate.

9.5 Is a volumetric capacity adequate?

The question is answered by proof: every load the volumetric benchmark dispatches is submitted to an oracle asking whether that exact set of RTIs can be stacked and placed in a trailer at all, with the true slot bound (46) in force so that an infeasible verdict is a proof. Of the 724 loads the benchmark dispatches over the thirty-six generated instances, 501 — 69.2% — are *proved* unplaceable. That figure is a floor and it is the one to read: 174 (24.0%) were left *undecided* within the limit and only 49 (6.8%) proved placeable,

so more oracle effort can only move loads out of the undecided column, and the net of the two pieces of evidence below is unknown. Re-running the same oracle at two and a half times the limit decided twenty more loads, all twenty placeable, leaving the proved count at 501; and the oracle carries no payload row, so an over-payload load comes back undecided rather than refuted. The share of *decided* loads that cannot be placed is correspondingly an upper reading, drifting downward with compute — 94.5% at the shorter limit, 91.1% at the longer — and rising to 91.8% at $v = 60$ where only 2.7% are undecided (Table 9).

The two ladder instances where the crush limit binds carry no undecided caveat at all: on $I_16_15_2_90_k25$, whose volumetric plan is the *proved* optimum, 41 of its 45 dispatched loads are proved unplaceable and four placeable, and on $I_24_20_2_90_k50$ it is 119 of 137. Those two went to the stronger oracle of Section 8.3 — a constructive witness first, the constraint programme only for refutations — which left nothing undecided; the thirty-six went to the constraint programme alone, which is why a quarter of their loads are. A volumetric optimum can therefore be a plan nine tenths of which cannot be loaded, and the error has a direction: a volumetric capacity ignores the remainders that footprints and layer heights leave unusable.

What that costs is constructed rather than estimated (Table 11): each volumetric plan's delivery quantities are kept and each period's freight re-packed with the same partition-and-pack machinery the method uses, every resulting truck verified. The re-packed plans need 74 and 212 trucks where the method plans 54 and 160 directly — savings of 27.0% and 24.5% on trucks and 19.1% and 16.2% on cost, each denominator the re-packed plan's own figure. Two caveats travel with them: the re-packed plan delivers *at least* the volumetric quantities, by 80 and 137 units, the partition being a cover; and the $I_24_20_2_90_k50$ volumetric plan is a memory stop rather than an optimum, so a better one might repair into fewer than 212 trucks and that row's margin is overstated in the method's favour. The $I_16_15_2_90_k25$ row, the proved optimum, carries no such caveat.

Deflating the capacity is the obvious remedy, and one step is measured: adding the stack-count rows (87) to the compact volumetric model, valid for every placeable load, raises its answer on $I_16_15_2_90_k25$ from forty-five trucks to forty-eight and its dual bound to 70 435.04 after 600 s — 2 296 above that model's own bound of 68 139.17 — while changing nothing where the crush limit does not bind. Whether those forty-eight loads can be placed has not been tested, the stack rows being necessary for placeability and not sufficient, so the stack-aware model is not yet a baseline.

The mechanism, and a test a planner can run. All of it has one cause, measured directly on $I_16_15_2_90_k25$: the payload admits 57 full RTIs of product P_1 in one truck while 32 is the largest number that can be placed, a proved bound from pricing that commodity alone to optimality. The container is 650 mm tall, so four layers fit under the 3 000 mm roof, while P_1 at 419 kg full fits only two under the 1 200 kg crush limit, as do six of the family's eight products. A volumetric fill therefore credits a truck with roughly twice that family's placeable count, which is also why the certified bound on the two crush-binding rungs sits 13.3% below the best verified plan.

That mechanism can be checked before anything is solved, since it needs only the catalogue and the trailer. For one container type, let the *analytic credit ratio* κ be the count a volumetric capacity credits over an *analytic* count of how many can be carried,

$$\kappa_r = \frac{\min\{\Gamma_K/\gamma_{pr},\ pK^w/\bar{w}_r\}}{\min\{s_r \min(L_r^h, L_r^c),\ pK^w/\bar{w}_r\}}, \qquad L_r^h = \left\lfloor \frac{rK}{r_{pr}} \right\rfloor, \quad L_r^c = \left\lfloor \frac{W^{\max}}{\bar{w}_r} \right\rfloor, \tag{88}$$

with $\bar{w}_r$ the catalogued gross weight of one full RTI and s_r the number of its footprints a single orientation tiles onto the floor. The denominator is a grid count with no packing oracle run, hence an *upper* estimate of what fits, so $\kappa > 1$ is a conservative diagnosis of over-credit. Over the fifteen types of the industrial catalogue κ has median 1.000, ranges to 2.000 and exceeds 1.20 on four types, and *no* type has $\kappa < 1$: the error cannot run in the planner's favour, and a plan over a mixed catalogue fails on its worst container. The census instances are *generated*, with κ in a range the generator controls, so this is a realism check on that range rather than a claim that the census shares transfer to a particular plant.

10 Conclusions

Procurement transport planning with returnable transport items has been formulated with transport capacity as what it physically is — a three-dimensional loading condition on every dispatched truck — and solved by branch and price over complete loaded trucks. Because 3D-SCPTOP is written over trucks it is block-angular, so the Dantzig–Wolfe master is obtained by aggregation and carries no loading configuration (Table 3), and the equivalence needs no hypothesis. Every lower bound reported rests on Proposition 2, which makes a bound affordable where a geometric certificate is not. And a volumetric transport capacity is

not a conservative simplification: the method plans the same quantities with 27.0% and 24.5% fewer trucks than re-packing two of its plans needs, and on the industrial catalogue the error runs one way only.

Three limits are real, and each names its next step. The method certifies no optimum here: the search ends with regions unrefuted, and the one optimum reported is established by inclusion from an exactly solved relaxation. The bound is loose exactly where the crush limit halves a container family's layer count, and the stack-count rows (87) tighten it measurably, so they are the candidate for tightening it inside the search as well, once it is known whether the loads they admit can be placed. Closing the regions the search leaves unrefuted may need a branching rule on *pairs* of columns, untested here. And the comparison against a procedure in industrial use is to be run on instances a plant solves itself, against a plan supplied to it.

A The SCPTOP benchmark

The volumetric benchmark extends the procurement transport model of (Díaz-Madroñero, Mula, and Peidro 2014) to the closed-loop two-node setting studied here: fuzzy parameters are dropped, transport capacity is expressed through the volume of the individual RTIs against the truck volume rather than a maximum container count, and a product may travel in more than one RTI type. To isolate the effect of the loading representation it is identical to the transport and inventory block of 3D-SCPTOP in every other respect — same two nodes, closed loop, lead time, coverage stock, payload cap, full-truckload floor and objective — and differs in one place: where 3D-SCPTOP asks whether the RTIs can be stacked and placed on the floor, SCPTOP asks only whether their volumes sum to less than the trailer's. The notation is that of Section 4, with $\overline{V}_{knt}$ a decision variable here rather than the slot sum (3).

$$\text{Min } z = \sum_{g\in G}\sum_{k\in K}\sum_{t\in T} cK\, Y_{gkt} + \sum_{g\in G}\sum_{n\in N}\sum_{t\in T} ic_{gn} I_{gnt} \tag{89}$$

subject to, for all $k \in K$ and $t \in T$,

$$\sum_{n\in N} \gamma_{p\rho(n)} \overline{V}_{knt} \le \Gamma_K\, Y_{pkt}, \qquad \sum_{r\in R} \gamma_{sr} V^{\mathrm{e}}_{krt} \le \Gamma_K\, Y_{skt} \tag{90}$$

$$\sum_{n\in N} w_n \overline{V}_{knt} \le pK^w Y_{pkt}, \qquad \sum_{r\in R} w^{\mathrm{f}}_r V^{\mathrm{e}}_{krt} \le pK^w Y_{skt} \tag{91}$$

together with the full-truckload floor on the binding resource, imposed exactly as in (52),

$$\sum_{n} \gamma_{p\rho(n)} \overline{V}_{knt} \;\ge\; \alpha\Gamma_K \big(Y_{pkt} - 1 + \theta_{pkt}\big), \qquad \sum_{n} w_n \overline{V}_{knt} \;\ge\; \alpha\, pK^w \big(Y_{pkt} - \theta_{pkt}\big), \tag{92}$$

with $\theta_{gkt} \in \{0, 1\}$ and the supplier direction analogous. The inventory balances, coverage stock and domains are (8)–(19) unchanged, and (45) is carried here too, so neither compact model is held to an easier standard. Constraint (90) is the whole of the benchmark's loading model: no footprint, stack height, layer count or crush limit appears.

B The thirty-six generated instances

Table 12 is Table 9 row by row.

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Table 12: The thirty-six generated instances through BP-3D-SCPTOP, every row on one commit, a 900 s budget, ten threads, a 20 GB per-model cap and seed 0, run sequentially. Wall times run from 603.1 s to 900.1 s; the longest row exceeds the budget by 0.1 s, because every phase of the construction carries a floor so that a phase begun just before the wall still gets a token solve. *ver* is the dispatched trucks an independent verifier re-derives and finds legal, out of the number dispatched; no plan here carries a complaint. LB is the certified volumetric relaxation, valid for 3D by Proposition 2 and printed only where every pair of the sweep came back proved. The last three columns are the volumetric benchmark SCPTOP on the same instance: the loads it dispatches, how many of them a constraint programme *proves* cannot be placed in a trailer at all, and how many it left *undecided* — which must be read beside the previous column, a low count of proven infeasibilities meaning nothing on a row where most loads were undecided. Row *I*_8_5_2_90 is the first rung of Table 7 under a *different run and commit*: the two agree on z, trucks and bound and differ by one node.

				BP-3D-SCPTOP						SCPTOP loads		
Instance	$\|N\|$	$\|T\|$	$\|R\|$	z	trucks	ver	LB (cert.)	gap	nodes	loads	unpl.	und.
*I*_4_10_2_120	4	10	2	62 505.08	45	45 / 45	53 425.21	14.53%	0	36	30	5
*I*_4_10_2_60	4	10	2	*no plan*	—	—	31 847.48	—	0	22	20	1
*I*_4_10_2_90	4	10	2	45 230.80	29	29 / 29	42 208.62	6.68%	3	26	22	3
*I*_4_10_3_120	4	10	3	57 953.00	42	42 / 42	47 690.39	17.71%	1	32	26	6
*I*_4_10_3_60	4	10	3	34 236.78	23	23 / 23	30 862.57	9.86%	2	20	18	2
*I*_4_10_3_90	4	10	3	44 485.13	30	30 / 30	44 456.13	0.07%	16	31	6	19
*I*_4_10_4_120	4	10	4	50 734.95	35	35 / 35	50 555.71	0.35%	4	35	12	21
*I*_4_10_4_60	4	10	4	*no plan*	—	—	35 872.62	—	0	24	23	0
*I*_4_10_4_90	4	10	4	43 698.86	29	29 / 29	40 487.62	7.35%	1	26	17	8
*I*_4_5_2_120	4	5	2	27 226.24	18	18 / 18	25 172.69	7.54%	10	16	16	0
*I*_4_5_2_60	4	5	2	14 715.58	9	9 / 9	13 768.49	6.44%	76	8	6	1
*I*_4_5_2_90	4	5	2	19 740.03	13	13 / 13	17 763.70	10.01%	126	11	9	2
*I*_4_5_3_120	4	5	3	25 925.74	16	16 / 16	25 787.76	0.53%	11	16	6	8
*I*_4_5_3_60	4	5	3	19 669.36	15	15 / 15	13 626.60	30.72%	1	9	9	0
*I*_4_5_3_90	4	5	3	21 502.06	14	14 / 14	21 498.70	0.02%	51	14	0	10
*I*_4_5_4_120	4	5	4	28 229.12	17	17 / 17	26 289.27	6.87%	19	15	9	5
*I*_4_5_4_60	4	5	4	16 166.20	12	12 / 12	12 969.54	19.77%	8	9	8	0
*I*_4_5_4_90	4	5	4	21 531.74	14	14 / 14	20 525.49	4.67%	19	13	7	5
*I*_8_10_2_120	8	10	2	63 265.31	44	44 / 44	55 483.16	12.30%	1	37	30	6
*I*_8_10_2_60	8	10	2	*no plan*	—	—	30 630.70	—	0	22	21	0
*I*_8_10_2_90	8	10	2	50 008.13	36	36 / 36	43 242.73	13.53%	1	30	27	1
*I*_8_10_3_120	8	10	3	58 632.27	40	40 / 40	52 387.71	10.65%	1	35	24	7
*I*_8_10_3_60	8	10	3	33 286.65	24	24 / 24	*unproved*	—	1	20	17	1
*I*_8_10_3_90	8	10	3	41 501.25	27	27 / 27	37 630.17	9.33%	1	24	21	2
*I*_8_10_4_120	8	10	4	66 209.55	44	44 / 44	58 184.10	12.12%	1	37	30	7
*I*_8_10_4_60	8	10	4	*no plan*	—	—	30 093.95	—	0	21	19	0
*I*_8_10_4_90	8	10	4	46 474.74	29	29 / 29	43 174.67	7.10%	1	27	15	11
*I*_8_5_2_120	8	5	2	26 181.24	16	16 / 16	26 066.68	0.44%	12	16	5	7
*I*_8_5_2_60	8	5	2	20 422.01	15	15 / 15	15 159.80	25.77%	3	10	9	0
*I*_8_5_2_90	8	5	2	18 796.96	12	12 / 12	18 795.54	0.01%	68	12	0	12
*I*_8_5_3_120	8	5	3	22 162.16	14	14 / 14	21 045.66	5.04%	2	13	9	3
*I*_8_5_3_60	8	5	3	*no plan*	—	—	14 101.19	—	0	9	9	0
*I*_8_5_3_90	8	5	3	18 649.29	12	12 / 12	18 648.12	0.01%	80	12	0	11
*I*_8_5_4_120	8	5	4	24 423.97	15	15 / 15	24 338.21	0.35%	8	15	5	7
*I*_8_5_4_60	8	5	4	17 693.47	12	12 / 12	13 578.09	23.26%	2	8	8	0
*I*_8_5_4_90	8	5	4	21 192.52	13	13 / 13	20 993.24	0.94%	6	13	8	3

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